Results Report: WT vs pifq single-cell RNA-seq pipeline

Executed results — real figures from a fresh run using the bibliography annotation (celltype); code shown as in the methods paper

SingleCell Pipeline

2026-07-28

Contents

Abstract **3**

Introduction **4**
About this document 4
Minimum computing requirements 5
Recommended: persistent terminal sessions with tmux or screen 5

1 Part 1 - Environment and raw data **7**
1.1 Clone the repository 7
1.2 Container startup (Docker or Singularity) 7
1.2.1 Which container runtime should I use? 8
1.3 Raw data download and Cell Ranger processing 9

2 Part 2 - Single-cell analysis in R **13**
2.1 Configuration 13
2.2 Initialization 13
2.3 Section 1 - Data loading and pre-filter QC 14
2.4 Section 2 - Cell filtering and doublet detection 16
2.5 Section 3 - Merge and initial preprocessing 18
2.6 Section 4 - Harmony integration 21
2.7 Section 5 - Choosing the clustering resolution 22
2.8 Section 6 - Final clustering 23
2.9 Section 7 - Cell-type annotation 25
2.10 Section 8 - Gene expression visualization 27
2.11 Section 9 - Export the annotated object 28

3 Part 3 - Pseudobulk differential expression and GO **29**
3.1 Section 10 - Cell-type subsets 29
3.2 Section 11 - Pseudo-replicate assignment 29
3.3 Section 12 - Pseudobulk tables and DESeq2 30
3.4 Section 13 - Volcano plots 31
3.5 Section 14 - Differential gene tables 32
3.6 Section 15 - GO enrichment 32
3.7 Section 16 - Log2FC heatmap 33
3.8 Section 17 - Gene-gene co-expression network (hdWGCNA) and TF network 35

4 Part 4 - Pseudotime trajectory analysis (Python) **37**
4.1 Section 18 - Setup 37

4.2 Section 19 - Load annotated object . . . 38
4.3 Section 20 - Cell type selection . . . 39
4.4 Section 21 - Trajectory inference . . . 41
4.5 Section 22 - Plot genes on trajectory . . . 43
4.6 Section 23 - Gene trends . . . 44
4.7 Section 24 - Final export . . . 44

Output layout **46**

Supplementary methods **47**
Supplementary Section S1 - Annotated clustree . . . 47
Supplementary Section S2 - Cell-type grouping . . . 47
Supplementary Section S3 - Subcluster inspection and manual curation . . . 48

Supplementary Table — Software dependencies and versions **51**
Interpreters and command-line tools . . . 51
Installing tmux or screen . . . 51
Installing a container runtime (Docker or Singularity/Apptainer) . . . 51
Troubleshooting Singularity/Apptainer . . . 52
R packages (421) . . . 52
Python packages (207) . . . 56

Abstract

A stepwise framework for inferring gene regulatory networks and pseudotime-based trajectories from single-cell RNA-sequencing data

Riveras E.[1], Vergara M.[1], Moreno-Ramirez S.[2], Gutiérrez R.A.[1]

[1]Departamento de Genética Molecular y Microbiología, Facultad de Ciencias Biológicas, Pontificia Universidad Católica de Chile, Santiago, Chile. Millennium Institute for Integrative Biology, Millennium Institute for Center for Genome Regulation. [2]Sainsbury Laboratory, University of Cambridge, Bateman Street, Cambridge CB2 1LR, UK.

Plant development is a dynamic process in which new organs are continuously formed throughout an individual's lifetime. This continuous production of new organs enables plants to increase in size and generate tissues with diverse physiological and functional capabilities. Single-cell RNA sequencing (scRNA-seq) has emerged as a powerful approach for characterizing cell-type-specific transcriptomes at high resolution across tissues and species. However, the bioinformatics analyses required to extract biological information from scRNA-seq datasets can be challenging for researchers without formal training in bioinformatics. Here, we provide a step-by-step workflow for inferring gene regulatory networks and pseudotime-based developmental trajectories from scRNA-seq data. To facilitate the implementation and reproducibility of these analyses, we provide a Docker image and a GitHub repository (github.com/laboratoriorgit-lang/SingleCell) containing all scripts, helper functions, reference files, container configuration, and environment YAML files. The workflow was applied to publicly available scRNA-seq data from *Arabidopsis thaliana* seedlings and enables the identification of differentially expressed genes (DEGs) across cell types, gene regulatory networks, and developmental trajectories associated with epidermal differentiation. Although this guide focuses on scRNA-seq analysis in *A. thaliana*, the workflow can be adapted to other organisms using appropriate reference genomes, genome annotations, and experimental designs.

Introduction

Plant organs are not laid down once during embryogenesis but are produced continuously across the plant's life, from meristems that generate new roots, leaves and flowers on demand. This ongoing organogenesis depends on precisely coordinated, cell-type-specific gene expression programs, as neighboring cells within the same tissue can follow markedly different transcriptional trajectories depending on their developmental origin, position, and stage of differentiation. Bulk RNA-seq, which averages expression across every cell in a sample, is poorly suited to resolving this heterogeneity: rare cell types are diluted out, and transitions between cell states are collapsed into a single average that may not represent any individual cell.

Single-cell RNA sequencing (scRNA-seq) addresses this limitation directly by profiling transcriptomes one cell at a time, making it possible to resolve cell-type identity, cell-state transitions, and developmental trajectories within a tissue at a resolution bulk approaches cannot reach. Once cells are clustered and annotated, the same data can support several complementary analyses: differential expression between cell types or genotypes, inference of gene regulatory networks from co-expression patterns, and reconstruction of pseudotime trajectories that order cells along a developmental or differentiation axis. In practice, however, each of these steps involves a distinct set of tools, file formats and statistical choices, and combining them into a coherent, reproducible pipeline requires computational expertise that is not always available in plant biology labs, which can be a real barrier to adopting scRNA-seq despite its potential.

This guide provides a complete, reproducible workflow that carries scRNA-seq data from raw sequencing reads through all of these downstream analyses. Starting from FASTQ files, Cell Ranger – a 10x Genomics program that aligns reads to a reference genome and counts, for every gene, how many transcripts were detected in each individual cell – is used to generate a per-cell count matrix (Part 1): this matrix (genes x cells) is the starting point for everything else in this workflow. These matrices are then loaded into R, where cells are filtered, doublets are removed, samples are merged and integrated with Harmony, and clusters are annotated to cell types using literature-curated marker genes (Part 2). From the annotated object, the pipeline builds pseudobulk profiles per cell type to identify differentially expressed genes between genotypes, runs GO enrichment on those gene sets, and infers a transcription-factor co-expression network to highlight candidate regulators (Part 3). Finally, the workflow switches to Python (scanpy, Palantir, scFates) to reconstruct pseudotime trajectories within a selected cell population, order cells along that trajectory, and identify genes whose expression trends define it (Part 4). Every step is packaged as a single container image, so the same software environment — R, Python, and every package version — is available to anyone who pulls the image, removing a major source of "works on my machine" irreproducibility.

To illustrate the complete workflow end to end, this document applies it to a public scRNA-seq dataset comparing wild-type and *pifq* mutant *Arabidopsis thaliana* seedlings, from raw Cell Ranger processing through cell-type annotation, differential expression, regulatory network inference, and a pseudotime trajectory of epidermal differentiation. Although the example organism is *A. thaliana*, none of the workflow logic is specific to it: the same steps apply to other species by substituting the reference genome, annotation, and marker gene set used for cell-type calling.

About this document

Important: this guide uses a subset of the published data. The study by Han et al. (2023) contains many more samples than the two analyzed here. This guide uses only two of them (`WT` and `pifq`) so that the whole workflow can be run in reasonable time on a single machine. It is **not** the same dataset as in the paper, so the clusters, differentially expressed genes, networks and trajectories obtained here are not expected to reproduce the paper's figures or numbers. The goal is to demonstrate the workflow, not to replicate the published results.

This methods-paper R Markdown documents the full WT vs pifq workflow: two samples, `WT` and `pifq`, processed and integrated together, then carried through three chapters — single-cell preprocessing, clustering and annotation (Part 2); pseudobulk differential expression, GO and the co-expression network (Part 3); and pseudotime trajectory analysis (Part 4). The code blocks are shown exactly as in the methods paper (not executed inline); the figures below are the **real outputs** of a fresh run of `workflow/step1_singlecell.R`, `workflow/step2_degs.R` and `workflow/step3_pseudotime.py`, written to a separate `results_rerun/` directory using the bibliography annotation (`celltype`).

The active annotation path is bibliography-based (`biblio_marks.txt -> celltype_curated`). Reference-transfer annotation is not part of this draft.

Minimum computing requirements

The table below summarizes the computing requirements for running the full pipeline on a personal computer, from raw FASTQ through the downstream Seurat/R and Python workflow. These are not hard limits; below them the pipeline still runs, just slower, and heavier steps (Cell Ranger, merging, PCA/UMAP) risk stalling or exhausting memory.

Component	Minimum	Recommended
CPU cores	4	16 or more
RAM	32 GB	64-128 GB for merged multi-sample objects
Disk space	80 GB free	100 GB free
Container runtime	Docker or Singularity/Apptainer (choose one; see “Container startup”)	Singularity/Apptainer on a cluster; Docker Desktop on a personal computer

This workflow is packaged as a single public image (`psblab/scrnaseq:latest`, on Docker Hub); the `docker-compose` YAML used to build and run it is kept in the repository. Any container runtime able to run that image works; the setup below covers both Docker and Singularity/Apptainer (the latter common on HPC clusters, where Docker is usually not permitted).

Small pilot datasets can be run on a laptop or desktop if memory is sufficient. For full experiments, repeated clustering sweeps, Cell Ranger processing from FASTQ, or large merged Seurat objects, a server or external cluster is strongly recommended. Cell Ranger is CPU- and memory-intensive and is not installed inside the downstream `psblab/scrnaseq:latest` Docker image.

Recommended: persistent terminal sessions with tmux or screen

Several steps in this pipeline run for a long time without any interaction — Cell Ranger takes 30-60 minutes per sample, and the R/Python clustering and pseudotime steps can run for hours on large merged objects. If these are launched directly in an SSH session and the connection drops (network hiccup, laptop sleep, closed terminal), the process is killed along with it.

Running everything inside a `tmux` or `screen` session avoids this: the session keeps running on the server after you disconnect, and can be reattached later — from the same machine or a different one — to check progress or pick up where things were left off. This is strongly recommended any time this pipeline runs on a remote server or HPC cluster, and is still useful on a local machine to keep a session alive while doing other work.

Either tool works; `tmux` is used in the examples below.

Bash

```bash
tmux new -s scrnaseq     # start a new named session
# ... run Cell Ranger, Docker/Singularity, R, etc. inside it ...
# detach without stopping anything: Ctrl-b, then d

tmux attach -t scrnaseq  # reattach later
tmux ls                  # list running sessions
```

This pipeline's R/Python environment ships as a single public container image (`psblab/scrnaseq:latest`, Docker or Singularity/Apptainer). It mounts the cloned repository as `/workspace`, so results are written back into the repo. Always pull the image rather than rebuilding it from the `Dockerfile`, so everyone gets the exact tested package versions (details and commands in "Container startup", Part 1).

Once inside a container (Docker or Singularity) shell, stay inside it for the rest of the work. The R and Python packages this pipeline needs (Seurat, scanpy, Palantir, scFates, etc.) are installed only inside the `psblab/scrnaseq:latest` image — they are not available on the bare host. Leaving the container shell (e.g. `exit`, or the shell dying) means the next command expecting those packages will fail. Pair this with tmux/screen: open a session, enter the container from inside it, and do the rest of the R/Python work (Parts 2-4) without leaving either until finished.

Installation instructions for tmux, screen, Docker and Singularity/Apptainer are in the Supplementary section at the end of this document.

1 Part 1 - Environment and raw data

1.1 Clone the repository

Choose or create a working directory, then clone the pipeline repository into it:

Bash

```bash
mkdir -p ~/my-analysis
cd ~/my-analysis
git clone https://github.com/laboratoriorgit-lang/SingleCell.git
cd SingleCell
```

All paths referenced later in this document (`workflow/`) are relative to this cloned repository. Its layout is:

```
SingleCell/
|-- Dockerfile
|-- docker-compose.yml
|-- README.md
|-- data/
|   |-- AtTFDB_loci.txt
|   |-- biblio_marks.txt
|   |-- Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta.gz
|   `-- Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf.gz
`-- workflow/
    |-- load_libraries.R                  # R packages
    |-- load_libraries_python.py          # Python packages
    |-- ScRNA_Analysis_Functions.R        # helper functions (Steps 1-2)
    |-- ScRNA_Pseudotime_Functions.py     # helper functions (Step 3)
    |-- step0_cellranger.sh               # Part 1 - Cell Ranger (bash)
    |-- step1_singlecell.R                # Part 2 - single-cell analysis
    |-- step2_degs.R                      # Part 3 - pseudobulk DE + GO
    `-- step3_pseudotime.py               # Part 4 - pseudotime (Python)
```

The bibliography marker table (`data/biblio_marks.txt`) is part of the repository for this WT vs pifq run (see Section 1).

1.2 Container startup (Docker or Singularity)

1.2.1 Which container runtime should I use?

You need **one** container program, not both. Docker and Singularity/Apptainer run the same image and give the same results; they differ only in where they can be installed and whether they need administrator rights. Use the table to pick one, then follow **only** the matching option below and skip the other.

Where you will run the pipeline	Use	Why
Institutional cluster or shared server (Linux)	**Option A: Singularity/Apptainer**	Needs no root; Docker is usually restricted to administrators
Personal Linux computer	**Option A: Singularity/Apptainer**, or **Option B: Docker** if it is already installed	Either works; reuse what you have
Personal macOS computer	**Option B: Docker** (Docker Desktop)	Apptainer does not run natively on macOS
Personal Windows computer	**Option B: Docker** (Docker Desktop with WSL2)	Apptainer does not run natively on Windows

A personal computer is only suitable for small pilot datasets (see the requirements above). For a full run, use a cluster or server and Option A.

Check what is already installed. Run these in a terminal; a version number means the program is available, and "command not found" means it is not:

Bash

```bash
apptainer --version        # Option A (newer name)
singularity --version      # Option A (older name; either one is enough)
docker --version           # Option B
docker ps                  # Option B only: an error here means you lack permission
```

On a cluster, Apptainer or Singularity is often installed system-wide (or available through `module avail apptainer` / `module load apptainer`). Docker may be installed but unusable if you are not in the `docker` group.

If neither is available:

- *Personal computer:* install the one chosen in the table above, following "Installing a container runtime" in the Supplementary section.
- *Cluster or shared server:* do not try to install anything. Ask the system administrators or support team, for example: "Please make Apptainer (Singularity) available so I can run the public image `docker://psblab/scrnaseq:latest`."

A container image is a single portable package holding a fixed, pre-installed set of R and Python libraries – exactly the versions this pipeline was tested with. Anyone who runs the same image gets an identical environment, with no version-mismatch problems across machines, operating systems, or clusters. The container mounts the cloned repository itself as `/workspace`, so results are written back into the repository (`results/`). Since results are not pushed upstream, this is not a problem. The same public image (`psblab/scrnaseq` on Docker Hub) runs under either runtime; always pull it rather than rebuilding from the `Dockerfile`, so every user gets the exact tested package versions.

Option A: Singularity / Apptainer. This is the container app used throughout this guide. No root is required, which is what makes it the practical choice on shared servers and HPC clusters (where Docker access is usually restricted to admins). It involves two separate steps:

1. **Build the `.sif` once.** This converts the public Docker Hub image into a local Singularity image file. Only needs to be done once per machine:

Bash

```
singularity build scrnaseq.sif docker://psblab/scrnaseq:latest
```

2. **Run it with `exec`.** This opens the working shell, binding the current directory to `/workspace` inside the container. Every time you come back to work on this pipeline, run this from inside the repository cloned in Section 1.1 (so `$(pwd)` resolves to the repository, not somewhere else):

Bash

```
singularity exec --bind $(pwd):/workspace --pwd /workspace scrnaseq.sif /bin/bash
```

If this step fails (for example with `squashfuse not found` or `Operation not permitted`), see “Troubleshooting Singularity/Apptainer” in the Supplementary section.

Apptainer imports the image environment, so the bundled Python (`/opt/venv`) and R stay on `PATH`, and it runs as the invoking user, so files under `results/` are owned by that user rather than by root. The `--pwd /workspace` flag starts the shell inside the bound repository (otherwise Apptainer warns that the host working directory does not exist in the container and falls back to `$HOME`).

Option B: Docker. Use this only if the table above sent you here, or if Singularity/Apptainer is not available. Requires root (or membership in the `docker` group), which is the reason it is not the default here. Mount the cloned repository and open a shell:

Bash

```
docker run -it --rm \
  -v $(pwd):/workspace \
  psblab/scrnaseq:latest /bin/bash
```

Or, from inside the cloned repository, `docker compose run r` uses the bundled `docker-compose.yml` (same image, same `/workspace` mount).

Stay inside this shell for Parts 2-4. All R and Python packages this pipeline needs are installed only inside this container image; running them outside of it fails. Start this shell inside a `tmux`/`screen` session (see above) so it survives a dropped connection.

1.3 Raw data download and Cell Ranger processing

Run this whole section inside the container, same as every other part of this pipeline: `cd` into the cloned repository and `singularity exec` into it (Section 1.2) before running any of the commands below. Write the Cell Ranger outputs into the cloned repository directory so they are visible from inside the container (which mounts that same directory as `/workspace`).

Now that the container is running, the first thing this pipeline needs is a count matrix: for every cell, how many transcripts of each gene were detected. That matrix is what the program **Cell Ranger** builds from the raw FASTQ reads, so it is the first tool run in this workflow, before anything in R or Python.

Install Cell Ranger 9.0.1 (proprietary 10x Genomics software; this workflow uses version 9.0.1). Get it from the 10x Genomics *previous versions* page:

`https://www.10xgenomics.com/support/software/cell-ranger/downloads/previous-versions`

1. Log in and accept the license.

2. Find **Cell Ranger 9.0.1** and copy its `curl` command.
3. That URL is signed and time-limited (it carries an `Expires=` timestamp tied to your account), so paste the current one — it cannot be hard-coded here or reused later. A plain `git clone` of the source repo does *not* give a ready-to-run binary.

Then extract it and add it to your `PATH`:

Bash

```bash
# Paste the signed curl command for 9.0.1 from the 10x page, e.g.:
curl -o cellranger-9.0.1.tar.gz "https://cf.10xgenomics.com/releases/cell-exp/\
cellranger-9.0.1.tar.gz?Expires=...&Key-Pair-Id=...&Signature=..."

tar -xzvf cellranger-9.0.1.tar.gz
export PATH=$PWD/cellranger-9.0.1/bin:$PATH
cellranger --version
```

Expected output: `tar` extracts a `cellranger-9.0.1/` folder; `cellranger --version` prints something like `cellranger cellranger-9.0.1` if the install and `PATH` are correct. If instead you get `command not found`, the `export PATH` line did not run, or was run in a different terminal/session than the one you are using now.

The single-cell workflow starts from Cell Ranger output, specifically the `outs/filtered_feature_bc_matrix/` folder for each sample. If only raw FASTQ files are available, first download or stage the FASTQs, then run Cell Ranger for each sample before starting the R workflow.

The WT vs pifq FASTQ files used in this guide are public data from Han et al., 2023, *Nature Plants*, deposited at the CNCB Genome Sequence Archive under BioProject PRJCA016521, run accession `CRA010863`. **Only two of the samples in that study are used here** (WT and pifq, listed below), not the full dataset, so results will differ from those reported in the paper:

Sample	Run accession	Files
ScWT	CRR775298	`CRR775298_f1.fastq.gz`, `CRR775298_r2.fastq.gz`
Scpifq	CRR775297	`CRR775297_f1.fastq.gz`, `CRR775297_r2.fastq.gz`

Download the reads, then rename/symlink them to the naming Cell Ranger expects (`<sample>_S1_L001_R1/R2_001.fastq.gz`)

Bash

```
mkdir -p /workspace/fastq/CRA010863

wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775298/CRR775298_f1.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775298/CRR775298_r2.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775297/CRR775297_f1.fastq.gz
wget -c -P /workspace/fastq/CRA010863 \
  ftp://download.big.ac.cn/gsa2/CRA010863/CRR775297/CRR775297_r2.fastq.gz

cd /workspace/fastq/CRA010863
ln -s CRR775298_f1.fastq.gz ScWT_S1_L001_R1_001.fastq.gz
ln -s CRR775298_r2.fastq.gz ScWT_S1_L001_R2_001.fastq.gz
ln -s CRR775297_f1.fastq.gz Scpifq_S1_L001_R1_001.fastq.gz
ln -s CRR775297_r2.fastq.gz Scpifq_S1_L001_R2_001.fastq.gz
```

Expected output: `ls /workspace/fastq/CRA010863` should show the four downloaded `.gz` files plus the four renamed symlinks (`ScWT_S1_L001_R1_001.fastq.gz`, etc.). Cell Ranger reads the symlink names, not the original ones, so if a rename step is skipped or misspelled, the next command below fails immediately with a “no input FASTQs found” error.

The reference genome (TAIR10.1) and Araport11 annotation ship compressed in the repository under `data/`, so no download is needed. Decompress them and build the Cell Ranger reference:

Bash

```
gunzip -k data/Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta.gz
gunzip -k data/Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf.gz

cellranger mkref \
  --genome=Ara \
  --fasta=data/Arabidopsis_thaliana_TAIR10.1_Araport11_genome.fasta \
  --genes=data/Arabidopsis_thaliana_TAIR10.1_Araport11_annotation.gtf
```

Expected output: this creates a folder named `Ara/` (matching `--genome`) containing the indexed reference Cell Ranger needs, and ends with a line reading `>>> Reference successfully created! <<<`. This only needs to be built once; the same `Ara/` folder is reused for every sample below.

Run Cell Ranger counts for each sample:

Bash

```
cellranger count --localcores=10 --id=ScWT --fastqs=/workspace/fastq/CRA010863 \
  --sample=ScWT --transcriptome=Ara --create-bam false

cellranger count --localcores=10 --id=Scpifq --fastqs=/workspace/fastq/CRA010863 \
  --sample=Scpifq --transcriptome=Ara --create-bam false
```

Expected output: each run creates a folder named after `--id` (`ScWT/`, `Scpifq/`) containing `outs/`, and ends with `Pipestance completed successfully!`. Inside `outs/`, `web_summary.html` is a self-contained QC report worth opening first (cell count, reads/cell, sequencing saturation); `filtered_feature_bc_matrix/` is the per-cell count matrix this whole pipeline is built on, and its path is what goes into the sample manifest in Section 1. This is the slowest step here, taking roughly 30-60 minutes per sample.

`--localcores=10` is used here as a portable default that keeps the example consistent across this report and the accompanying workflow scripts; it is not a hard requirement. Cell Ranger will happily use more or fewer cores, so raise or lower this value to match the number of CPU cores actually available on the machine running the pipeline (see the Recommended column above for guidance).

Check the Cell Ranger output before moving on. A poor run is much easier to diagnose here than after clustering. Open `ScWT/outs/web_summary.html` and `Scpifq/outs/web_summary.html` in a browser (the same numbers are in `outs/metrics_summary.csv`) and look at the following:

Metric	What it tells you	What to look for
Estimated number of cells	Barcodes called as cells	Close to the number of cells you intended to capture; very different from the expected value suggests a loading or calling problem
Mean reads per cell / median genes per cell	Sequencing depth and per-cell complexity	Both should be of the same order in the two samples; a large imbalance can introduce a technical difference between WT and pifq
Sequencing saturation	How much deeper sequencing would still add new transcripts	Low values mean more sequencing would recover more genes; very high values mean little is gained
Valid barcodes and reads in cells	Library quality and how much signal is in real cells versus ambient background	Low values indicate a poor library or a high background
Reads mapped confidently to the transcriptome	Reference suitability	A low fraction can mean a wrong or incomplete reference or annotation

The **barcode rank plot** (the “knee plot”, under the *Cells* panel) ranks barcodes by total UMI counts. A good run shows a sharp drop (the “knee”) separating a high-count group of cells from a long tail of empty droplets. A gradual slope with no clear knee, or a knee at a very different position in the two samples, means that cell calling was uncertain and the cell counts of the two samples may not be comparable. These values are not strict pass/fail thresholds, and they depend on the tissue and the protocol; use them to compare the two samples with each other and to spot samples that look clearly different from the other.

Cell Ranger’s cell calling is only a first pass: the QC filtering in Part 2 (Section 2) removes low-quality cells and doublets from this set, so the cell number reported here is expected to be larger than the one used downstream.

2 Part 2 - Single-cell analysis in R

From this point on, the pipeline runs in R inside the container (Singularity). Start R:

Bash

```bash
R
```

The next two steps, Configuration and Initialization, set up the paths and helper functions used by every section that follows.

2.1 Configuration

This block defines the main paths used by the pipeline: the working directory, the folder where the workflow scripts are stored, and the folder where all results will be written. Edit these values before running the analysis if your project is mounted in a different location.

R

```r
PIPELINE_DIR <- "/workspace/workflow"
DATA_DIR    <- "/workspace/."
base_dir    <- file.path(DATA_DIR, "results")
```

2.2 Initialization

This block prepares the R session before running the analysis. It loads the pipeline helper scripts, sets a fixed random seed for reproducibility, configures Seurat compatibility options, moves R into the project working directory, and creates the output folders used by the rest of the workflow. The seed is important because some analysis steps include random components; keeping it fixed helps obtain comparable results when the pipeline is rerun.

R

```r
source(file.path(PIPELINE_DIR, "load_libraries.R"))
source(file.path(PIPELINE_DIR, "ScRNA_Analysis_Functions.R"))
set.seed(1807)
options(Seurat.allow.s4 = FALSE)
setwd(DATA_DIR)
list2env(create_pipeline_dirs(base_dir), envir = .GlobalEnv)
output_dir <- base_dir
```

After this block runs, the R session is ready to execute the analysis sections. The helper functions are available, the project paths are fixed, and the results folders have been created. Each following section only needs to set `output_dir` to the corresponding folder before saving figures or objects.

2.3 Section 1 - Data loading and pre-filter QC

This section loads each sample from the Cell Ranger output and creates the first quality-control plots before filtering.

To use this block, edit the sample list with one entry per sample:

- `file`: path to the `filtered_feature_bc_matrix` folder, relative to `DATA_DIR`.
- `label`: short sample name used in plots and object labels.
- `condition`: experimental group or treatment condition.

Then edit `colors`, assigning one color to each sample `label`. These colors are used in the QC plots and later visualizations.

The mitochondrial gene pattern tells the pipeline how to identify mitochondrial reads. For Arabidopsis, mitochondrial genes usually start with `ATMG`; this is the default used in this report. The resulting mitochondrial percentage (`percent.mt`) is used to filter cells in Section 2.

After running the block, the pipeline creates one raw Seurat object per sample and saves the pre-filter QC plot in `results/01_qc/qc_prefilter.pdf`. This plot is used to inspect sample quality before choosing filtering thresholds in the next section.

R

```r
output_dir <- dir_01
samples <- list(
  list(
    file      = "ScWT/outs/filtered_feature_bc_matrix",
    label     = "WT",
    condition = "WT"
  ),
  list(
    file      = "Scpifq/outs/filtered_feature_bc_matrix",
    label     = "pifq",
    condition = "pifq"
  )
)
colors <- c(
  "WT"   = "#66c2a5",
  "pifq"   = "#fc8d62"
)
mt_pattern <- "^ATMG"
seurat_list_raw <- load_seurat_samples(samples = samples,
                                       DATA_DIR = DATA_DIR,
                                       mt_pattern = mt_pattern)
plot_qc_batch(seurat_list_raw, colors, "qc_prefilter.pdf")
message("\nOK SECTION 1 COMPLETE: QC pre-filter plots saved")
```

Representative output from Section 1:

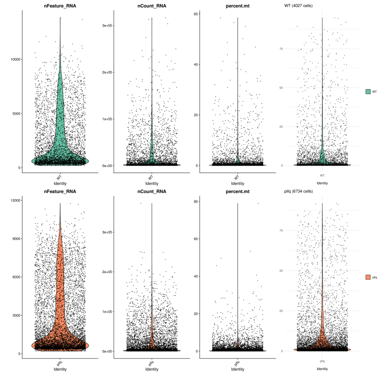


Figure 1: Pre-filter QC violin plots.

2.4 Section 2 - Cell filtering and doublet detection

This section filters low-quality cells, then runs **DoubletFinder** to flag and remove doublets (droplets that captured two cells instead of one, which would otherwise show up as a fake intermediate or hybrid cell type). The code shown here uses two filtering thresholds – cells must have more than 200 detected genes (`min_features = 200`) and less than 5% mitochondrial reads (`max_mt = 5`) – and DoubletFinder runs automatically after that filtering step, since `run_doubletfinder` defaults to `TRUE` in `filter_sample()`; it is not something that needs to be added, only disabled if you don't want it (see below). Internally, this calls `doubletfinder_pipeline()`, which normalizes, scales, runs PCA, sweeps DoubletFinder's `pK` parameter, and removes any cells it classifies as doublets.

The filtering function also accepts extra options that are not used in this example (which filters only by `min_features = 200` and `max_mt = 5`). Their defaults do not filter anything:

Option	Default	When to use it	How to choose a value
`max_features`	`Inf`	Cells with far more detected genes than the rest, often doublets	In the pre-filter violin of `nFeature_RNA`, cut where the upper tail separates from the main body of cells (for example, near the 99th percentile)
`min_counts`	`0`	Cells with very few UMIs even though they pass `min_features`	Same approach on `nCount_RNA`, at the lower end
`max_counts`	`Inf`	Cells with far more UMIs than the rest	Same approach on `nCount_RNA`, at the upper end
`run_doubletfinder`	`TRUE`	Set to `FALSE` only to skip doublet detection in a quick test run	Not applicable

After running the block, the pipeline saves a post-filter QC plot in `results/01_qc/qc_postfilter.pdf` and stores the filtered sample list in `results/objects/seurat_list_postfilter.rds`. This saved object is used by the next section.

R

```r
output_dir <- dir_01
seurat_list <- filter_seurat_samples(
  seurat_list_raw, min_features = 200, max_mt = 5, run_doubletfinder = TRUE
)
plot_qc_batch(seurat_list, colors, "qc_postfilter.pdf")
saveRDS(seurat_list, file.path(dir_objects, "seurat_list_postfilter.rds"))
message("\nOK SECTION 2 COMPLETE: filtering and doublet detection complete")
```

Representative output from Section 2:

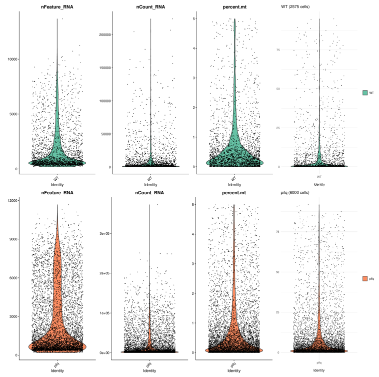


Figure 2: Post-filter QC violin plots.

2.5 Section 3 - Merge and initial preprocessing

This section merges the two filtered samples into a single Seurat object and runs the standard preprocessing steps on it. The result is a merged object that has **not** yet been integrated across samples (that is Section 4).

What the data look like. At this point every sample is a matrix with one row per gene and one column per cell; each entry is the number of transcripts (UMIs) detected for that gene in that cell. Most entries are zero, so Seurat stores the matrix in a sparse format and prints zeros as `.`. You can inspect the top-left corner of the WT matrix yourself (6 genes by 4 cells):

R

```r
GetAssayData(seurat_list[["WT"]], layer = "counts")[1:6, 1:4]
```

Expected output:

```
6 x 4 sparse Matrix of class "dgCMatrix"
          AAACCCAGTGATAGAT-1 AAACCCATCAGTGCGC-1 AAACGAAAGGCAGCTA-1 AAACGAACAAGCGGAT-1
AT1G01010                  .                  .                  .                  .
AT1G01020                  .                  .                  .                  1
AT1G01030                  .                  .                  .                  .
AT1G01040                  .                  .                  .                  .
AT1G01050                  1                  .                  .                  .
AT1G01060                  2                  .                  .                  .
```

Each row is a gene (TAIR identifier) and each column is a cell (identified by its barcode, with the suffix `-1` added by Cell Ranger). A number is how many transcripts of that gene were detected in that cell: for example, `AT1G01060` has 2 in the first cell. A dot means zero. Even in this small corner most entries are zero, which is typical of single-cell data and the reason the sparse format is used.

Everything below transforms this matrix. The steps are chained in one pipe, in this order:

1. **Merge** (`reduce(seurat_list, merge)`) joins the samples into one object that keeps the sample of origin of each cell in `orig.ident`.
2. **Normalize** (`NormalizeData`) divides each cell's counts by its total and multiplies by 10,000, then log-transforms the result (`LogNormalize`). This makes cells with different sequencing depth comparable.
3. **Select highly variable genes (HVGs)** (`FindVariableFeatures`) keeps the 2,000 genes (`nfeatures = 2000`) whose expression varies most across cells after accounting for their average level (`selection.method = "vst"`). Most genes are expressed similarly in all cells and add mostly noise; the variable ones carry the differences between cell types, so the next steps use only them. This is the feature selection step of the workflow.
4. **Scale** (`ScaleData`) centers and scales each selected gene to mean 0 and variance 1 (values are capped at 10), so that highly expressed genes do not dominate the next step. By default only the HVGs are scaled.
5. **PCA** (`RunPCA(npcs = 30)`) summarizes the scaled HVGs into 30 principal components (PCs), new axes that capture the main sources of variation. The later steps work on these PCs instead of on thousands of genes.
6. **UMAP** (`RunUMAP(dims = 1:30)`) projects those 30 PCs into two dimensions for visualization. It is useful to see how the samples look before Harmony.

Number of PCs. All 30 PCs are used in every later step (Harmony, UMAP and clustering use `dims = 1:30`).

Default parameters used here. Several important settings are not written in the code because they are Seurat defaults (Seurat 5.5.1). Change them if your data call for it:

Step	Parameter	Value	What it does
`NormalizeData`	`scale.factor`	10,000	Total counts per cell after normalization, before the log transform
`FindVariableFeatures`	`nfeatures`	2,000	Number of HVGs kept (set explicitly here)
`ScaleData`	`scale.max`	10	Cap on scaled values, to limit the effect of outliers
`RunPCA`	`npcs`	30	Number of PCs computed (set explicitly here)
`RunUMAP`	`n.neighbors`	30	Neighbors used to build the map; lower values emphasize local structure
`RunUMAP`	`min.dist`	0.3	How tightly cells are allowed to pack together in the plot
`FindNeighbors` (Section 6)	`k.param`	20	Neighbors per cell in the graph used for clustering

The output is saved as `results/objects/ath_sc_preharmony.rds`, and the UMAP before Harmony integration is saved as `results/01_qc/umap_preharmony.pdf`.

R

```r
output_dir <- dir_01
ath_sc <- reduce(seurat_list, merge) %>%
  NormalizeData(verbose = FALSE) %>%
  FindVariableFeatures(
    selection.method = "vst", nfeatures = 2000, verbose = FALSE
  ) %>%
  ScaleData(verbose = FALSE) %>%
  RunPCA(npcs = 30, verbose = FALSE) %>%
  RunUMAP(reduction = "pca", dims = 1:30, verbose = FALSE)
save_pdf(
  DimPlot(ath_sc, group.by = "orig.ident", cols = colors),
  "umap_preharmony.pdf"
)
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_preharmony.rds"))
message("\nOK SECTION 3 COMPLETE: merge and preprocessing complete")
```

Representative output from Section 3:

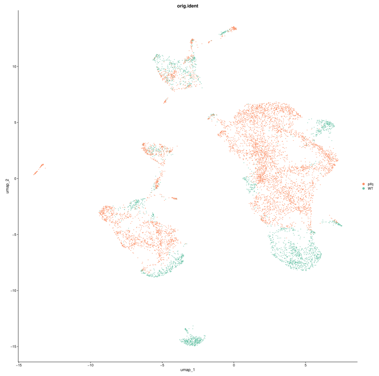


Figure 3: UMAP before Harmony correction, colored by sample.

2.6 Section 4 - Harmony integration

The two samples in this guide are different genotypes (WT and pifq), so they differ for biological reasons as well as for technical ones. Left as they are, cells can cluster by *sample* instead of by *cell type*, which would hide the cell populations that both samples share. This section applies **Harmony** to reduce those sample-associated differences in the PCA representation calculated in Section 3, so that cells of the same type from both samples are placed together. The correction is done using the sample identity stored in `orig.ident`.

Harmony is one of several integration methods (others include Seurat's CCA/RPCA integration, fastMNN, Scanorama and scVI); we use it here because it is fast, works directly on the PCA embedding and is widely used. Harmony only changes the embedding used for UMAP and clustering; it does not modify the gene counts. Because each genotype is a single sample here, Harmony cannot tell technical differences from genotype effects, so check that the biological differences you care about are still visible after correction.

After Harmony, the UMAP is recalculated using the corrected representation. This second UMAP helps check whether samples mix better while still preserving biological structure.

The corrected object is saved as `results/objects/ath_sc_postharmony.rds`, and the post-Harmony UMAP is saved as `results/01_qc/umap_postharmony.pdf`.

R

```r
output_dir <- dir_01
ath_sc <- ath_sc %>%
  RunHarmony("orig.ident", plot_convergence = FALSE) %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE)
save_pdf(
  DimPlot(ath_sc, group.by = "orig.ident", cols = colors),
  "umap_postharmony.pdf"
)
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_postharmony.rds"))
message("\nOK SECTION 4 COMPLETE: Harmony integration complete")
```

Representative output from Section 4:

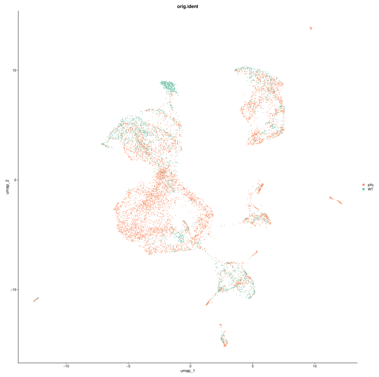


Figure 4: UMAP after Harmony correction, colored by sample.

2.7 Section 5 - Choosing the clustering resolution

This section compares several candidate clustering resolutions with a **clustree**. The clustree shows, for each resolution in `resolutions_test`, how many clusters it produces and whether cells stay in stable clusters or keep moving between them as the resolution changes. A resolution where cell membership stops moving is a safer choice than one that is still fragmenting. Lower resolutions produce fewer, broader clusters; higher resolutions produce more, smaller ones. The values in `resolutions_test` are only the range of candidates to compare; widen or narrow it if the useful region lies outside it.

The output is saved in `results/02_clustering/` as `clustree2.pdf`. Use it to choose `cluster_resolution` in Section 6.

R

```r
resolutions_test <- c(0.15, 0.35, 0.45, 0.55, 1.0)
output_dir <- dir_02
clu <- run_resolution_sweep(
  ath_sc, resolutions = resolutions_test, output_dir = output_dir
)
message("\nOK SECTION 5 COMPLETE: clustree saved")
```

Representative output from Section 5:

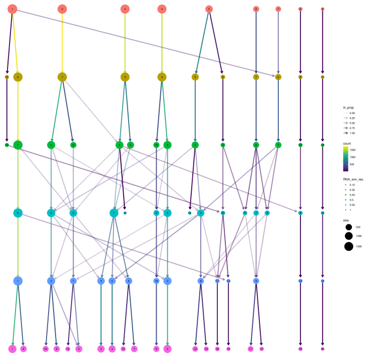


Figure 5: Cluster tree across candidate Leiden resolutions.

2.8 Section 6 - Final clustering

This section groups cells that look transcriptionally similar into discrete clusters, using the **Leiden algorithm** (`algorithm = 4`) – a standard community-detection method for single-cell data. Each cluster is a candidate cell population; cell-type labels are only assigned afterward, in Section 7. The resolution is set in `cluster_resolution`, chosen from the clustree in Section 5. Lower resolutions produce fewer, broader clusters; higher resolutions produce more, smaller ones. Clustering uses the first 30 Harmony dimensions (`dims = 1:30`).

The resolution used here is `cluster_resolution <- 0.35`. In the clustree (Section 5), 0.35 is one of the most stable resolutions: its clusters mostly trace back to a single cluster of the previous resolution (solid arrows). Raising the resolution beyond it (0.45 and above) starts to split clusters into many smaller ones and cells begin to move between branches (the pale crossing arrows), which would give clusters that are harder to interpret as cell types. This is a judgment call, not a fixed rule: with another dataset, choose the resolution where the clustree is stable and the number of clusters still matches the cell populations you expect.

`FindNeighbors` builds a graph that connects each cell to its `k.param = 20` nearest neighbors in Harmony space, and `FindClusters` then splits that graph into communities. A smaller `k.param` gives finer, more local clusters; a larger one gives smoother, broader ones.

After running this block, each cell receives a final Seurat cluster ID stored in `seurat_clusters`. These numeric clusters are the starting point for cell-type annotation in the next section.

The output figure is a UMAP colored by final cluster number, saved as `results/02_clustering/umap_seuratclusters.pdf`.

R

```r
cluster_resolution <- 0.35
output_dir <- dir_02
ath_sc <- ath_sc %>%
  RunUMAP(reduction = "harmony", dims = 1:30, verbose = FALSE) %>%
  FindNeighbors(
    reduction = "harmony", dims = 1:30, k.param = 20, verbose = FALSE
  ) %>%
  FindClusters(resolution = cluster_resolution, algorithm = 4, verbose = FALSE)
Idents(ath_sc) <- "seurat_clusters"
save_pdf(
  DimPlot(ath_sc, group.by = "seurat_clusters", label = TRUE),
  "umap_seuratclusters.pdf"
)
message("\nOK SECTION 6 COMPLETE: final clustering complete")
```

Representative output from Section 6:

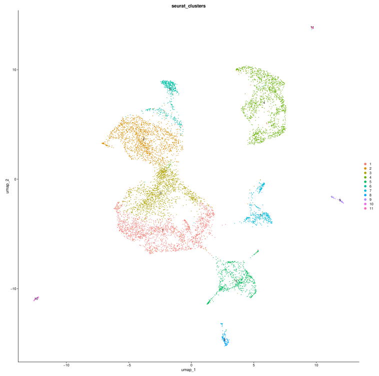


Figure 6: Final UMAP colored by numbered Seurat clusters.

2.9 Section 7 - Cell-type annotation

This section assigns biological cell-type labels to the clusters using a marker gene table from the literature. The marker file is read from `data/biblio_marks.txt`.

The pipeline first finds marker genes for the dataset (recomputed each time with `force = TRUE`, so they always match the current clusters), then compares those markers with the bibliography table to assign a `celltype` label. The dotplot is used to inspect whether the expected marker genes support the assigned labels.

Marker table format. The marker table has one row per marker gene and three columns: `celltype` (the cell type the gene marks), `symbol` (the gene name) and `ID` (the gene identifier, the AGI code for Arabidopsis, which must match the gene names of the Seurat object). An optional `source` column can record where each marker comes from; the code ignores it. The first rows of the table look like this:

```
celltype        symbol   ID
guard cell      FAMA     AT3G24140
guard cell      TGG1     AT5G26000
guard cell      EPF1     AT2G20875
companion cell  AHA3     AT5G57350
companion cell  APL      AT1G79430
companion cell  PP2-A1   AT4G19840
```

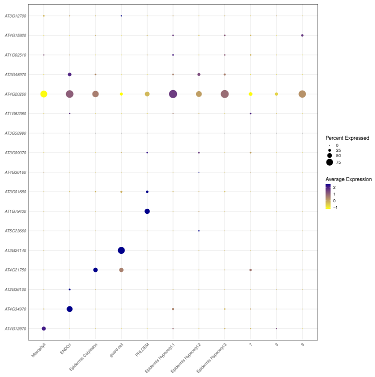
The table is read with `standardize_marker_table()`, which finds the columns by their header names. An older two-column table (cell type and gene ID) is still accepted; in that case no symbol is available and the plots show the ID. In the dotplot, genes are labeled with their symbol by default; set `name_gene = FALSE` to label them with the ID only.

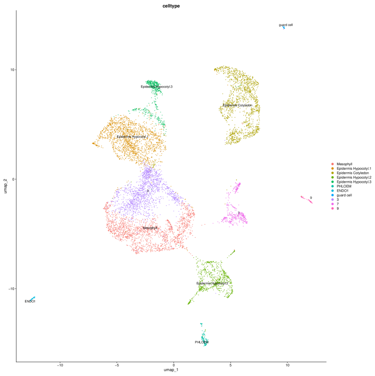
The marker table is saved in `results/03_annotation/FindAllMarkers.tsv`, and the main figures are saved as `dotplot_marker_table_annotation_biblio.pdf` and `umap_annotation_biblio.pdf`.

R

```r
output_dir <- dir_03
biblio_marks_file <- file.path(DATA_DIR, "data", "biblio_marks.txt")
marker_table <- standardize_marker_table(biblio_marks_file)
markers <- find_markers(
  ath_sc,
  output_file = file.path(output_dir, "FindAllMarkers.tsv"),
  force = TRUE
)
ath_sc <- annotate_by_markers(
  ath_sc, markers,
  reference_file = biblio_marks_file
)
plot_marker_dotplot(
  ath_sc,
  marker_table,
  annot_col = "celltype",
  name_gene = TRUE,
  outfile   = file.path(
    output_dir, "dotplot_marker_table_annotation_biblio.pdf"
  ),
  width = 18, height = 18
)
save_pdf(
  DimPlot(
    ath_sc, group.by = "celltype",
    label = TRUE, repel = TRUE, raster = FALSE
  ),
  "umap_annotation_biblio.pdf"
)
message("\nOK SECTION 7 COMPLETE: cell-type annotation complete")
```

Representative outputs from Section 7:





2.10 Section 8 - Gene expression visualization

This section plots the expression of one selected gene across the annotated cell types. The gene to inspect is set in the `gene` variable.

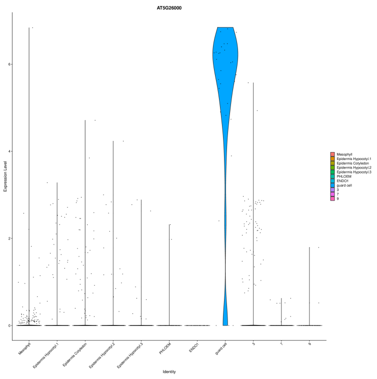
The violin plot shows expression levels by cell type, while the FeaturePlot shows where the same gene is expressed on the UMAP. These two views are useful for checking whether a marker or gene of interest matches the expected cell population.

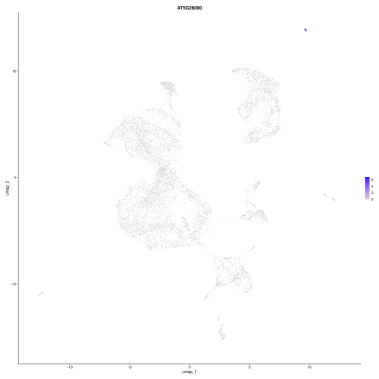
The outputs are saved in `results/04_expression/` with filenames based on the selected gene.

R

```r
gene <- "AT5G26000"
output_dir <- dir_04
ath_sc <- JoinLayers(ath_sc)
Idents(ath_sc) <- "celltype"
gene_tag <- paste(gene, collapse = "_")
save_vln(
  VlnPlot(ath_sc, features = gene),
  paste0("vln_gene_", gene_tag, ".pdf")
)
save_pdf(
  FeaturePlot(ath_sc, features = gene),
  paste0("feature_gene_", gene_tag, ".pdf")
)
message("\nOK SECTION 8 COMPLETE: gene expression visualization saved")
```

Representative outputs from Section 8:





2.11 Section 9 - Export the annotated object

This section closes the R part of the workflow by saving the final object in two formats. The `.rds` file is used to continue working in R, and the `.h5ad` file is used by the Python sections of the pipeline.

The outputs are saved in `results/objects/` as `ath_sc_annotated.rds` and `ath_sc_annotated.h5ad`.

R

```r
# Checkpoint - restore with:
#   ath_sc <- readRDS(file.path(dir_objects, "ath_sc_annotated.rds"))
saveRDS(ath_sc, file.path(dir_objects, "ath_sc_annotated.rds"))

# Export the annotated object to AnnData h5ad format for Python-based
# trajectory and velocity analyses.
export_to_scanpy(
  ath_sc,
  file.path(dir_objects, "ath_sc_annotated.h5ad")
)

# To export a specific cell type only:
# export_to_scanpy(
#   subset(ath_sc, subset = celltype == "guard cell"),
#   file.path(dir_objects, "GuardCell.h5ad")
# )

message("\nOK SECTION 9 COMPLETE: annotated object saved (.rds + .h5ad)")
```

3 Part 3 - Pseudobulk differential expression and GO

This part continues from the object saved at the end of Part 1. It aggregates cells into pseudo-replicates per cell type, runs pseudobulk DESeq2 for the WT vs pifq contrast, and summarizes results with volcano plots, differential tables, GO enrichment, a log2FC heatmap, and co-expression networks.

R

```r
ath_sc <- readRDS(file.path(dir_objects, "ath_sc_annotated.rds"))
```

3.1 Section 10 - Cell-type subsets

The annotation column used to define cell types is a user choice. Depending on how Part 1 was run, this is one of `celltype` (bibliography annotation), `celltype_grouped` (broad groups from Supplementary Section S2), or `celltype_curated` (manual curation from Supplementary Section S3). All downstream sections in this part use whichever column is set here.

R

```r
pseudobulk_annot_col <- "celltype"
table(ath_sc[[pseudobulk_annot_col]])
cell_type_subsets <- create_cell_type_subsets(
  ath_sc, annot_col = pseudobulk_annot_col
)
message("\nOK SECTION 10 COMPLETE: cell-type subsets created")
```

Cells per cell type (bibliography annotation `celltype`; the three epidermis hypocotyl subclusters and the still-numeric unannotated clusters are kept separate at this stage):

Cell type	Cells
Mesophyll	1739
Epidermis Cotyledon	1569
Epidermis Hypocotyl.1	1691
Epidermis Hypocotyl.2	752
Epidermis Hypocotyl.3	498
PHLOEM	127
ENDO1	67
guard cell	31
3	1586
7	437
9	78

3.2 Section 11 - Pseudo-replicate assignment

This section creates pseudo-replicates for the pseudobulk analysis. Instead of treating each single cell as an independent replicate, cells from the same cell type and condition are split into a small number of groups. Each group becomes one pseudo-replicate.

The number of pseudo-replicates is controlled by `n_pseudoreps`. In this report, `n_pseudoreps = 3`, so each condition inside each cell type is split into three pseudo-replicates when enough cells are available.

`pseudobulk_conditions` can be left as `NULL` to use all conditions present in the object, or it can be set to specific conditions if only some groups should be used. Cell types that do not contain enough information for a comparison, such as cell types present in only one condition, are removed from the pseudobulk contrast.

The output is `cell_type_subsets_replicates`, which is the same set of cell-type subsets but with pseudo-replicate labels added. This object is used in the next section to build pseudobulk count tables.

R

```r
pseudobulk_conditions <- NULL
n_pseudoreps          <- 3
cell_type_subsets_replicates <- assign_pseudoreplicates_batch(
  cell_type_subsets,
  pseudobulk_conditions = pseudobulk_conditions,
  n_pseudoreps          = n_pseudoreps
)
table(cell_type_subsets_replicates[["Mesophyll"]]$replicate)
message("\nOK SECTION 11 COMPLETE: pseudo-replicates assigned")
```

Cell types present in only one condition (here cluster `9`, with only 78 cells) are dropped, since a WT vs pifq contrast needs both conditions represented. Ten cell types are retained.

Cells per pseudo-replicate (first two cell types shown):

Cell type	Replicate	Cells
Mesophyll	pifq_rep1	499
Mesophyll	pifq_rep2	527
Mesophyll	pifq_rep3	488
Mesophyll	WT_rep1	72
Mesophyll	WT_rep2	76
Mesophyll	WT_rep3	77
Epidermis Cotyledon	pifq_rep1	…
Epidermis Cotyledon	pifq_rep2	…
…	…	…

3.3 Section 12 - Pseudobulk tables and DESeq2

Testing every single cell as its own replicate overstates statistical confidence, since cells from the same sample are not independent. This section avoids that by building pseudobulk count tables and running **DESeq2** – the standard, widely-cited tool for differential expression from count data – for the selected comparisons. For each cell type, counts from cells belonging to the same pseudo-replicate are summed together, creating replicate-level count profiles that can be analyzed the same way a bulk RNA-seq experiment would be.

The comparisons are defined in the `comparisons` list. Each entry contains the two conditions to compare and a short `tag` used to name the output folder and files. In this report, the comparison is `WT` versus `pifq`, and the tag is `WT_vs_pifq`.

`cell_types_to_analyze` can be left as `NULL` to analyze all available cell types, or set to a selected list if only specific cell types should be tested.

For each cell type and comparison, the pipeline writes a DESeq2 result table with one row per gene. The main columns are `log2FoldChange`, `pvalue`, and `padj`. For the `WT_vs_pifq` comparison used here, positive log2FC means higher expression in `pifq`, while negative log2FC means higher expression in `WT`.

R

```
comparisons <- list(
  list(conds = c("WT", "pifq"), tag = "WT_vs_pifq")
)
cell_types_to_analyze <- NULL
output_dir <- dir_06
deseq2_results <- run_pseudobulk_deseq2_analysis(
  cell_type_subsets_replicates = cell_type_subsets_replicates,
  comparisons    = comparisons,
  output_dir     = output_dir,
  cell_types     = cell_types_to_analyze,
  pseudobulk_dir = file.path(dir_objects, "pseudobulk_replicates")
)
message("\nOK SECTION 12 COMPLETE: pseudobulk aggregation and DESeq2 complete")
```

One `DESeq2_<cell_type>_WT_vs_pifq.csv` is written per cell type: the standard DESeq2 results table (one row per gene, with `log2FoldChange`, `pvalue`, and `padj`). Positive log2FC means higher expression in pifq. First rows of the Mesophyll table:

```
             baseMean  log2FoldChange   lfcSE     stat    pvalue      padj
AT1G01010     94.04    -1.659           0.294   -5.652   1.6e-08     1.3e-07
AT1G01020    119.88     0.204           0.168    1.212   2.3e-01     3.6e-01
AT1G01030    417.65    -0.605           0.172   -3.521   4.3e-04     1.8e-03
```

3.4 Section 13 - Volcano plots

This section creates volcano plots from the DESeq2 result tables generated in Section 15. Each plot summarizes one cell type for the selected comparison.

The `volcano_tag` selects which comparison folder to use. The thresholds `padj_cut` and `lfc_cut` define which genes are highlighted as significant in the plots. In this report, genes are highlighted when adjusted p-value is below 0.05 and absolute log2FC is at least 1.

The volcano plots are saved inside the corresponding comparison folder under `results/06_de_results/`.

R

```
volcano_tag <- "WT_vs_pifq"
padj_cut    <- 0.05
lfc_cut     <- 1
render_volcano_plots(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag, "volcano"),
  pdf_name    = paste0("VolcanoPlots_", volcano_tag, ".pdf"),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut
)
message("\nOK SECTION 13 COMPLETE: volcano plots saved")
```

Representative output from Section 13:

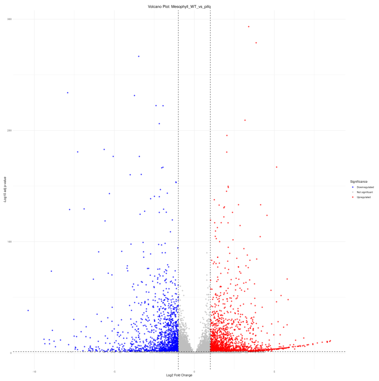


Figure 7: Volcano plot for Mesophyll, WT vs pifq.

3.5 Section 14 - Differential gene tables

This section summarizes the DESeq2 results into combined differential gene tables. It uses the same significance thresholds defined for the volcano plots: `padj_cut` for adjusted p-value and `lfc_cut` for log2 fold-change.

The output makes it easier to compare differential expression across cell types. One table records whether each gene is significantly upregulated, downregulated, or not significant in each cell type. A second table stores the corresponding log2FC values for the significant genes.

These tables are useful for quickly checking which genes are shared across cell types and which responses are cell-type specific.

R

```r
diff_prefix <- paste0("diff_table_", volcano_tag)
diff_tables <- build_differential_tables(
  results_dir = file.path(dir_06, volcano_tag),
  output_dir  = file.path(dir_06, volcano_tag),
  padj_cut    = padj_cut,
  lfc_cut     = lfc_cut,
  prefix      = diff_prefix
)
message("\nOK SECTION 14 COMPLETE: differential gene tables saved")
```

This writes two combined matrices (genes x cell types). The class matrix (`diff_table_WT_vs_pifq_*.tsv`) uses `+1/-1/0` to flag genes up-, down-, or not significantly regulated in each cell type; the log2FC matrix (`log2FC_table_*.tsv`) holds the actual fold changes (`NA` where not significant). Example rows for three annotated cell types (Epidermis Cotyledon, Mesophyll, PHLOEM):

```
# class matrix                          # log2FC matrix
gene_id     EpiCot  Meso  Phlo           gene_id     EpiCot    Meso    Phlo
AT1G01010     -1     -1     0           AT1G01010    -1.90    -1.66     NA
AT1G01060      1      1     0           AT1G01060     1.47     4.55     NA
AT1G01120     -1      1     0           AT1G01120    -1.72     1.22     NA
```

3.6 Section 15 - GO enrichment

This section tests whether the significant genes from each cell type are enriched for Gene Ontology terms. In simple terms, it asks whether the genes that change between conditions are concentrated in particular biological processes, functions, or cellular components.

The `go_space` variable selects which GO category to use. Here it is set to `BP`, meaning Biological Process. The organism database is set with `go_orgdb`, and `go_keytype` tells the function which gene identifier format is being used. For this Arabidopsis report, the database is `org.At.tair.db` and the key type is `TAIR`.

The adjusted p-value cutoff is controlled by `padj_cutoff`. The GO result tables and bubble plots are saved in `results/07_go/` under the selected comparison name.

R

```r
go_space    <- "BP"
go_orgdb    <- org.At.tair.db
go_keytype  <- "TAIR"
padj_cutoff <- 0.05
go_results <- run_go_enrichment_for_contrast(
  results_dir  = file.path(dir_06, volcano_tag),
  output_dir   = file.path(dir_07, volcano_tag),
  orgdb        = go_orgdb,
  keytype      = go_keytype,
  go_space     = go_space,
  padj_cutoff  = padj_cutoff,
  contrast_tag = volcano_tag
)
message("\nOK SECTION 15 COMPLETE: GO enrichment complete")
```

Representative output from Section 15:

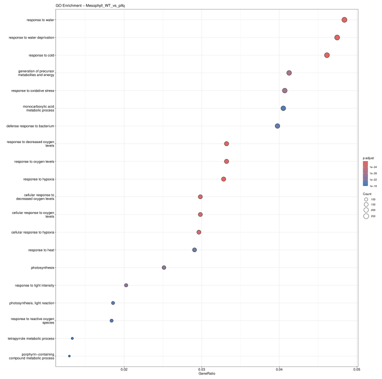


Figure 8: GO enrichment (BP) for Mesophyll, WT vs pifq.

3.7 Section 16 - Log2FC heatmap

This section turns the differential expression tables into a heatmap of log2FC values. It is used to compare how strongly genes change across cell types in the same condition contrast.

Rows are genes and columns are cell types. Genes with similar cross-cell-type patterns are grouped together. The `heatmap_limits` value controls the color scale; in this report it is set from -5 to 5 so strong negative and positive changes are shown on the same scale.

The heatmap is saved in the comparison folder inside `results/06_de_results/`.

R

```r
heatmap_limits <- c(-5, 5)
build_logfc_heatmap(
  logfc_table  = diff_tables$logfc,
  contrast_tag = volcano_tag,
  output_dir   = file.path(dir_06, volcano_tag),
  limits       = heatmap_limits,
  cell_order   = unique(marker_table$cell.types)
)
message("\nOK SECTION 16 COMPLETE: log2FC heatmap saved")
```

Representative output from Section 16:

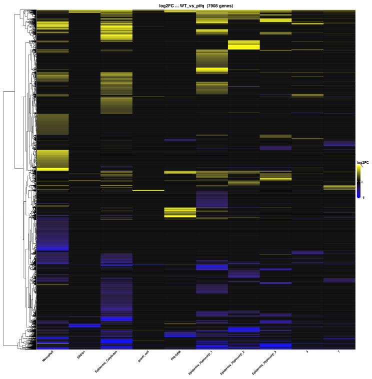


Figure 9: log2FC heatmap: rows hierarchically clustered, columns in marker-table order (unannotated clusters last), WT vs pifq.

3.8 Section 17 - Gene-gene co-expression network (hdWGCNA) and TF network

This section builds a gene-gene co-expression network with **hdWGCNA** (high-dimensional weighted gene co-expression network analysis; Morabito et al., 2023, *Cell Reports Methods*), the single-cell version of WGCNA (Langfelder and Horvath, 2008, *BMC Bioinformatics*). The network is built from the significant differentially expressed genes and then focused on transcription factor (TF) connections. The TF list comes from `data/AtTFDB_loci.txt`, which was prepared from AtTFDB, the Arabidopsis transcription factor database available through AGRIS: https://agris-knowledgebase.org/.

hdWGCNA works in four steps. It groups similar cells into **metacells** to reduce the sparsity of single-cell counts, chooses a **soft-thresholding power** that gives a network with a scale-free-like structure, computes the **topological overlap matrix (TOM)** between genes, and detects **modules** of co-expressed genes. The TOM value is the weight of each gene-gene edge.

The main input is the log2FC table produced in Section 17. The wrapper function `run_hdwgcna_tf_network()` (its name says which method it runs) takes the Seurat object, the differential expression results and the annotation column, runs hdWGCNA, keeps the strong gene-gene edges, and draws the TF network colored by differential expression direction.

The parameters near the end of the call control practical details: the number of metacells, the minimum module size, the edge-strength threshold (a minimum TOM weight) and how many hub TFs are labeled. The network output is saved in `results/08_networks/`.

R

```r
run_hdwgcna_tf_network(
  seurat_obj      = ath_sc,
  de_table_path   = file.path(dir_06, volcano_tag, "log2FC_table_fc1_padj_005.tsv"),
  tf_list_path    = file.path(DATA_DIR, "data", "AtTFDB_loci.txt"),
  output_dir      = dir_08,
  annot_col       = pseudobulk_annot_col,
  contrast_tag    = volcano_tag,
  n_metacells     = 50,
  min_module_size = 20,
  deep_split      = 2,
  soft_power      = NULL,
  tom_threshold   = 0.2,
  n_hub_label     = 15
)
message("\nOK SECTION 17 COMPLETE: TF coexpression network saved")
```

Representative output from Section 17:

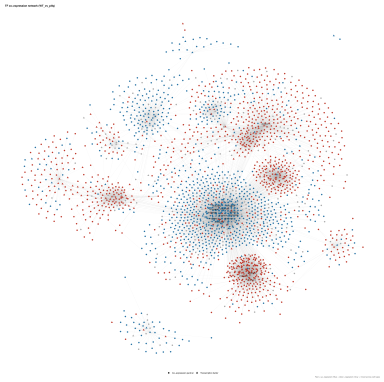

TF co-expression network (WT_vs_pifq)
● Co-expression partner
▲ Transcription factor
Red = up-regulated | Blue = down-regulated | Gray = mixed across cell types

Figure 10: TF co-expression network, colored by DE direction.

4 Part 4 - Pseudotime trajectory analysis (Python)

Part 4 runs in Python (scanpy, Palantir, scFates), starting from the h5ad object exported at the end of Part 1 (`results/objects/ath_sc_annotated.h5ad`). Exit R and start Python inside the same container:

Bash

```
q()          # leave R (do not save the workspace)
python3
```

4.1 Section 18 - Setup

This section starts the Python part of the pipeline. It defines the same project paths used earlier, loads the Python helper scripts, and creates the pseudotime output folder.

Use this block first after leaving R and starting Python inside the container. After it runs, the functions needed for trajectory analysis are available in the Python session.

Python

```
import os
PIPELINE_DIR = "/workspace/workflow"
DATA_DIR     = "/workspace/."
base_dir     = os.path.join(DATA_DIR, "results")
exec(open(os.path.join(PIPELINE_DIR, "load_libraries_python.py")).read(), globals())
exec(open(os.path.join(PIPELINE_DIR, "ScRNA_Pseudotime_Functions.py")).read(), globals())
dir_pseudotime = os.path.join(base_dir, "09_pseudotime")
os.makedirs(dir_pseudotime, exist_ok=True)
print("SECTION 18 COMPLETE: setup done")
```

4.2 Section 19 - Load annotated object

This section loads the `.h5ad` object exported at the end of the R workflow. The `INPUT_H5AD` path points to that file, and `ANNOTATION_COL` selects which cell type annotation column will be used for plotting and selecting cells.

Up to three annotation columns can be present, depending on which optional (supplementary) steps were run in Step 1:

- `celltype` – the direct output of Section 7 (bibliography-marker annotation). The rawest of the three, and always present.
- `celltype_grouped` – present only if Supplementary Section S2 was run. Same as `celltype`, but with related subclusters merged into a single label (e.g. the two epidermis cotyledon subclusters collapsed into one).
- `celltype_curated` – present only if Supplementary Section S3 was run. The manually curated annotation, built by inspecting and relabeling subclusters.

Set `ANNOTATION_COL` to whichever of these matches how far the optional steps went in Step 1: `celltype` if Supplementary Sections S2 and S3 were skipped, `celltype_grouped` if only Supplementary Section S2 was used, and `celltype_curated` if Supplementary Section S3 was used.

The example below uses `celltype`, the rawest of the three. Under this column, epidermis cotyledon is split into `Epidermis Cotyledon.1` and `.2` – both options for handling this are shown in Sections 20-21 below.

The function also standardizes coordinate names so the object works cleanly with Scanpy. It prints the annotation columns available in the loaded object – use this to confirm `ANNOTATION_COL` is set correctly – then saves an overview UMAP and prints the available cell types, which are used to choose the trajectory cells in the next section.

Python

```python
INPUT_H5AD     = "/workspace/results/objects/ath_sc_annotated.h5ad"
ANNOTATION_COL = "celltype"
N_JOBS         = 4
adata, N_JOBS = load_curated_object(
    input_h5ad     = INPUT_H5AD,
    dir_pseudotime = dir_pseudotime,
    annotation_col = ANNOTATION_COL,
    n_jobs         = N_JOBS,
)
```

Representative output from Section 19:

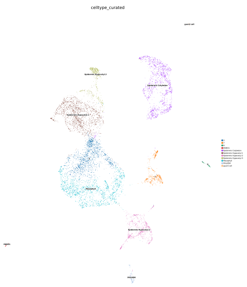


Figure 11: Overview UMAP colored by annotated cell type.

4.3 Section 20 - Cell type selection

This section selects which cell types will be used for trajectory inference. The list is set in `TRAJECTORY_CLUSTERS`, and the names must match the cell-type labels printed in Section 24.

In this report, only epidermis cotyledon is selected. The preview UMAP shows these cells before fitting the trajectory, which helps confirm that the intended population was chosen.

The exact label(s) to use here depend on which `ANNOTATION_COL` was set in Section 19:

- With `ANNOTATION_COL = "celltype"` (raw), the two cotyledon subclusters are told apart and both must be listed.
- With `ANNOTATION_COL = "celltype_grouped"` (Supplementary Section S2), all cotyledon cells are already one population.

Tip: if the goal is to evaluate the trajectory across the whole cell type without splitting it into subclusters, `celltype_grouped` is the better choice here.

Option A – with `ANNOTATION_COL = "celltype"` (raw, subclusters told apart):

Python

```python
TRAJECTORY_CLUSTERS = ["Epidermis Cotyledon.1", "Epidermis Cotyledon.2"]
adata_sub = preview_trajectory_selection(
    adata          = adata,
    clusters       = TRAJECTORY_CLUSTERS,
    annotation_col = ANNOTATION_COL,
    dir_pseudotime = dir_pseudotime,
)
print("SECTION 20 COMPLETE: cell type selection saved")
```

Option B – with `ANNOTATION_COL = "celltype_grouped"` (all cotyledon cells as one population; preferred when subclusters should not be told apart):

Python

```python
TRAJECTORY_CLUSTERS = ["Epidermis Cotyledon"]
adata_sub = preview_trajectory_selection(
    adata          = adata,
    clusters       = TRAJECTORY_CLUSTERS,
    annotation_col = ANNOTATION_COL,
    dir_pseudotime = dir_pseudotime,
)
print("SECTION 20 COMPLETE: cell type selection saved")
```

Representative output from Section 20:

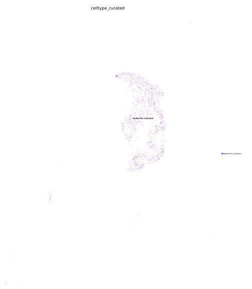


Figure 12: UMAP of the epidermis cotyledon cells selected for the trajectory.

4.4 Section 21 - Trajectory inference

This section fits the pseudotime trajectory for the selected cells. The `ROOT_CLUSTER` defines where pseudotime starts, and must be a single existing label (not a list): with raw `celltype` this means picking one of the two cotyledon subclusters (the larger one, `.1`, is used below); with `celltype_grouped` it is simply the merged `Epidermis Cotyledon` label.

`TRAJECTORY_RUNS` stores the trajectory settings to test. Each run creates its own output folder, making it possible to compare parameter choices without overwriting previous results. The run shown here uses 50 nodes, `sigma = 0.3`, `lambda_value = 200`, `eigs = 3`, and `seed = 3`.

Here, `eigs = 3` means that three diffusion components are used before fitting the trajectory. The outputs include the trajectory graph, pseudotime visualization, and selected root cell plot inside `results/09_pseudotime/`.

Option A – with `ANNOTATION_COL = "celltype"` (raw, pick one of the two cotyledon subclusters as root – the larger one, `.1`, is used here):

Python

```python
ROOT_CLUSTER = "Epidermis Cotyledon.1"
TRAJECTORY_RUNS = [
    trajectory_run(nodes=50, sigma=0.3, lambda_value=200, eigs=3, seed=3),
]
adata_traj, selected_trajectory_dir, trajectory_runs = run_trajectory_runs(
    adata           = adata,
    clusters        = TRAJECTORY_CLUSTERS,
    root_cluster    = ROOT_CLUSTER,
    annotation_col  = ANNOTATION_COL,
    output_base_dir = dir_pseudotime,
    runs            = TRAJECTORY_RUNS,
)
print("SECTION 21 COMPLETE: trajectory inference saved")
```

Option B – with `ANNOTATION_COL = "celltype_grouped"` (preferred when evaluating the trajectory across the whole cell type):

Python

```python
ROOT_CLUSTER = "Epidermis Cotyledon"
TRAJECTORY_RUNS = [
    trajectory_run(nodes=50, sigma=0.3, lambda_value=200, eigs=3, seed=3),
]
adata_traj, selected_trajectory_dir, trajectory_runs = run_trajectory_runs(
    adata           = adata,
    clusters        = TRAJECTORY_CLUSTERS,
    root_cluster    = ROOT_CLUSTER,
    annotation_col  = ANNOTATION_COL,
    output_base_dir = dir_pseudotime,
    runs            = TRAJECTORY_RUNS,
)
print("SECTION 21 COMPLETE: trajectory inference saved")
```

Representative outputs from Section 21 for the epidermis-cotyledon-only trajectory using `nodes=50`, `sigma=0.3`, `lambda=200`, and `eigs=3`:

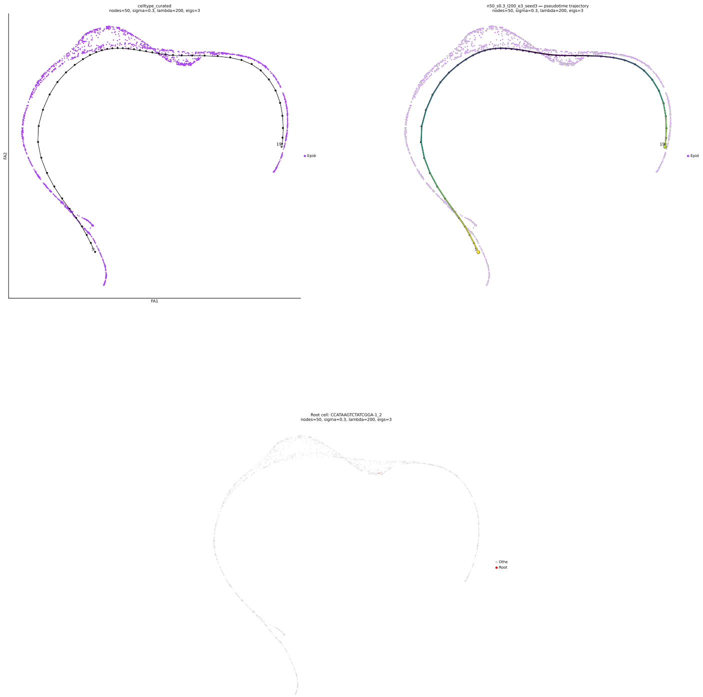


Figure 13: Root cell (red) automatically selected within the epidermis cotyledon population as the most-connected cell — pseudotime = 0 here.

4.5 Section 22 - Plot genes on trajectory

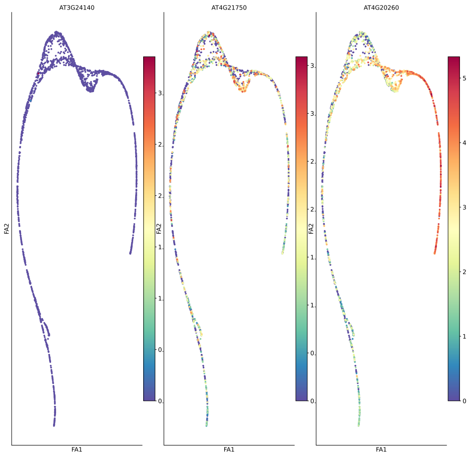
This section plots selected genes on the fitted epidermis cotyledon trajectory. The gene list is controlled by `GENES`, so changing that vector is the only thing needed to inspect another marker.

For this version, Section 22 is focused only on ATML1. In the count matrix ATML1 is stored as the locus `AT4G21750`, so the code uses `AT4G21750` while the text and figure label report it as ATML1. The plot shows where ATML1 expression falls on the epidermis cotyledon trajectory layout.

Python

```python
GENES = ["AT4G21750"]
name  = os.path.basename(selected_trajectory_dir)
ensure_expression_in_X(adata_traj)
gene_plots_dir = os.path.join(selected_trajectory_dir, "gene_plots")
os.makedirs(gene_plots_dir, exist_ok=True)
fig = sc.pl.draw_graph(
    adata_traj,
    color=GENES,
    title=["ATML1 (AT4G21750)"],
    show=False,
    return_fig=True,
)
save_plot_18x18(fig, os.path.join(gene_plots_dir, f"{name}_gene_plots.pdf"))
plt.close(fig)
print("SECTION 22 COMPLETE: gene trajectory plots saved")
```

Representative output from Section 22:



ATML1 (`AT4G21750`) expression on the epidermis cotyledon trajectory.

4.6 Section 23 - Gene trends

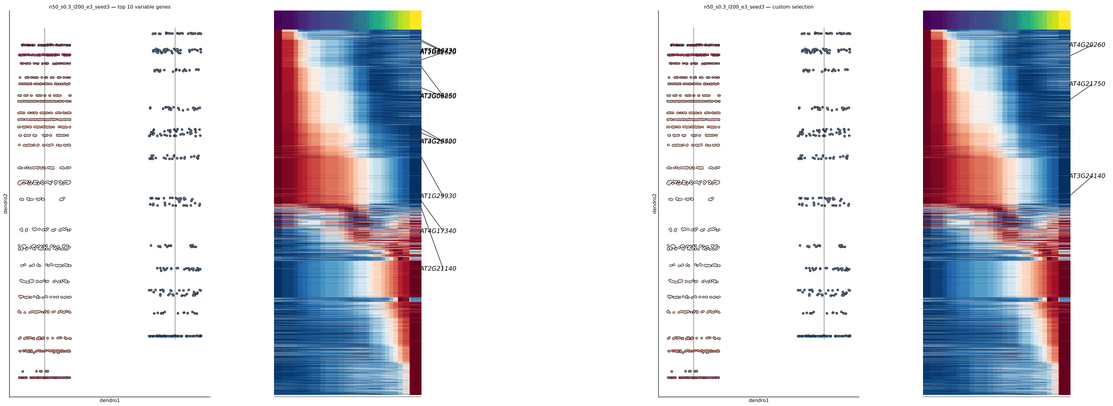
This section fits gene-expression trends along pseudotime for the epidermis cotyledon trajectory calculated above. The function first tests genes for association with pseudotime, fits smooth expression curves, ranks genes by their trend signal, and then saves trend heatmaps.

For this report, the run shown here uses the already selected epidermis cotyledon trajectory: `nodes=50`, `sigma=0.3`, `lambda=200`, `eigs=3`, and `seed=3`. `TOP_N = 10` means the main panel shows the top 10 ranked genes. `HIGHLIGHT_GENES = ["AT3G24140"]` highlights the selected marker using the same fitted trend workflow.

Python

```python
TOP_N           = 10
HIGHLIGHT_GENES = ["AT3G24140"]
ORDERING        = "max"
SPLINE_DF       = 3
name = os.path.basename(selected_trajectory_dir)
ensure_expression_in_X(adata_traj)
top_path, highlight_path = run_step29_gene_trends(
    adata         = adata_traj,
    run_dir       = selected_trajectory_dir,
    name          = name,
    custom_genes  = HIGHLIGHT_GENES,
    top_n         = TOP_N,
    ordering      = ORDERING,
    n_jobs        = N_JOBS,
    spline_df     = SPLINE_DF,
)
print("SECTION 23 COMPLETE: epidermis cotyledon gene trends saved")
```

Representative outputs from Section 23 — epidermis cotyledon gene trends:



Left: top 10 ranked genes along pseudotime for the epidermis cotyledon trajectory. Right: `AT3G24140` highlighted from the same fitted trend workflow.

4.7 Section 24 - Final export

This section saves the final pseudotime object after trajectory inference and gene-trend analysis. The exported `.h5ad` file keeps the trajectory results, pseudotime values, and layout coordinates for later reuse.

The final object is saved in `results/objects/` as `ath_sc_pseudotime_final.h5ad`.

Python

```
final_h5ad = os.path.join(base_dir, "objects", "ath_sc_pseudotime_final.h5ad")
os.makedirs(os.path.dirname(final_h5ad), exist_ok=True)
adata_traj.write_h5ad(final_h5ad)
print(f"SECTION 24 COMPLETE: final pseudotime object saved -> {final_h5ad}")
```

This is the last step of the pipeline. Leave the Python REPL the same way Part 4 started – by exiting it explicitly, mirroring `q()` on the R side:

Bash

```
exit()        # leave Python
```

Output layout

All four parts write into the same **`results/`** folder inside the cloned repository (mounted as **`/workspace/results`** in the container), one subfolder per step. After running the full pipeline once, it looks like this:

```
results/
|-- 01_qc/
|   |-- qc_prefilter.pdf
|   |-- qc_postfilter.pdf
|   |-- umap_preharmony.pdf
|   `-- umap_postharmony.pdf
|-- 02_clustering/
|   |-- clustree2.pdf
|   `-- umap_seuratclusters.pdf
|-- 03_annotation/
|   |-- FindAllMarkers.tsv
|   |-- dotplot_marker_table_annotation_biblio.pdf
|   |-- umap_annotation_biblio.pdf
|   `-- clustree_annotated.pdf
|-- 04_expression/
|   |-- feature_gene_<gene>.pdf
|   `-- vln_gene_<gene>.pdf
|-- 05_curation/
|   |-- umap_grouped.pdf
|   |-- subcluster_markers_<cluster>.pdf
|   `-- umap_curated.pdf
|-- objects/                              # checkpoints (.rds / .h5ad)
|   |-- seurat_list_postfilter.rds
|   |-- ath_sc_preharmony.rds
|   |-- ath_sc_postharmony.rds
|   |-- ath_sc_annotated.rds
|   |-- ath_sc_annotated.h5ad
|   |-- pseudobulk_replicates/
|   `-- ath_sc_pseudotime_final.h5ad
|-- 06_de_results/<comparison_tag>/       # e.g. WT_vs_pifq/
|   |-- DESeq2_<cell_type>_<tag>.csv
|   |-- <cell_type>_<tag>_filtered.csv
|   |-- diff_table_<tag>_fc1_padj_005.tsv
|   |-- log2FC_table_fc1_padj_005.tsv
|   |-- heatmap_<tag>.pdf
|   `-- volcano/
|       |-- DESeq2_<cell_type>_<tag>.png
|       `-- VolcanoPlots_<tag>.pdf
|-- 07_go/<comparison_tag>/
|   |-- GO_<cell_type>_<tag>_BP.tsv
|   `-- GO_<cell_type>_<tag>_BP_bubble.pdf
|-- 08_networks/
|   |-- edges_unified.tsv
|   |-- nodes_unified.tsv
|   |-- modules_unified.tsv
|   |-- hubgenes_unified.tsv
|   |-- hdwgcna_summary.tsv
|   |-- unified_TOM.rda
|   |-- hdwgcna_unified.rds
|   `-- network_tf_DE_direction.pdf
`-- 09_pseudotime/
    |-- umap_overview.pdf
    |-- umap_selection.pdf
    `-- trajectory/<run_name>/            # e.g. n50_s0.3_l200_e3_seed3/
        |-- parameters.tsv
        |-- <run_name>_annotation.pdf
        |-- <run_name>_pseudotime_trajectory.pdf
        |-- <run_name>_root_cell.pdf
        |-- <run_name>_pseudotime_by_cell.tsv
        |-- <run_name>_trajectory.h5ad
        |-- gene_plots/<run_name>_gene_plots.pdf
        `-- gene_trends/
            |-- step29_gene_trends_top10_FINAL_24x18.pdf
            `-- step29_gene_trends_highlight_list_FINAL_24x18.pdf
```

`01_qc`-`05_curation` and `objects/` come from Part 2 (`step1_singlecell.R`); `06_de_results`/`07_go` (one subfolder per `comparisons` tag, Section 12) and `08_networks` from Part 3 (`step2_degs.R`); `09_pseudotime` from Part 4 (`step3_pseudotime.py`), one `trajectory/<run_name>/` per entry in `TRAJECTORY_RUNS` (Section 21). `<gene>`, `<cluster>`, `<cell_type>`, and `<tag>` depend on the values used when running each section, not on fixed names.

Supplementary methods

The three steps below are optional. They start from the annotation of Section 7 and write their results to separate columns (`celltype_grouped`, `celltype_curated`) or figures, so the main workflow runs without them. If you use one of them, set the annotation column of Supplementary Section S2 accordingly. The code is also available in `workflow/step1_supplementary.R`.

Supplementary Section S1 - Annotated clustree

This section adds the cell-type labels from Section 7 onto the clustering tree from Section 5. Instead of showing only numeric clusters, each node is labeled with the most common cell type found in that cluster.

This makes it easier to see how stable the annotated cell types are across different clustering resolutions. The output is saved as `results/03_annotation/clustree_annotated.pdf`.

R

```
output_dir <- dir_03
plot_annotated_clustree(
  ath_sc, clu, annot_col = "celltype", output_dir = output_dir
)
message("\nOK SUPPLEMENTARY S1 COMPLETE: annotated clustree saved")
```

Representative output from Supplementary Section S1:

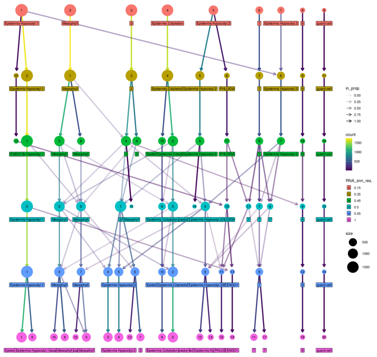


Figure 14: Cluster tree labeled by the dominant bibliography-based annotation in each node.

Supplementary Section S2 - Cell-type grouping

This optional section simplifies annotation labels by merging selected cell types or subclusters into broader groups. It uses the `grouping` object as a small renaming table: the names on the left are the current labels, and the values on the right are the new labels.

For example, in this report the two mesophyll subclusters, `Mesophyll.1` and `.2`, are renamed as `Mesophyll`. Any cell type that is not listed in `grouping` keeps its original label.

Use this section when the detailed annotation is useful for inspection but a simpler grouping is better for figures or downstream summaries. The grouped labels are stored in `celltype_grouped`, and the grouped UMAP is saved as `results/05_curation/umap_grouped.pdf`.

R

```r
# Merge the two mesophyll subclusters into a single label
grouping <- c(
  "Mesophyll.1" = "Mesophyll",
  "Mesophyll.2" = "Mesophyll"
)

output_dir <- dir_05

if (length(grouping) > 0) {
  current <- as.character(ath_sc$celltype)
  ath_sc$celltype_grouped <- ifelse(current %in% names(grouping), grouping[current], current)
} else {
  ath_sc$celltype_grouped <- ath_sc$celltype
}

save_pdf(
  DimPlot(
    ath_sc, group.by = "celltype_grouped",
    label = TRUE, repel = TRUE, raster = FALSE
  ),
  "umap_grouped.pdf"
)

message("\nOK SUPPLEMENTARY S2 COMPLETE: cell-type grouping complete")
```

Representative output from Supplementary Section S2:

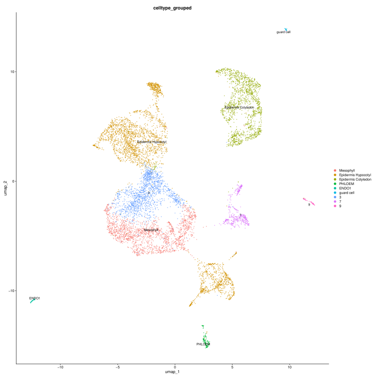


Figure 15: UMAP colored by grouped cell-type labels.

Supplementary Section S3 - Subcluster inspection and manual curation

This optional section is for manual curation when one annotated cluster needs a closer look. First, `inspect_subcluster_markers()` takes one cluster, splits it into smaller subclusters, and saves a figure with the subcluster UMAP and marker gene patterns. This helps decide whether the original label should be kept or refined.

If refinement is needed, `reassign` is edited as a small mapping table. The outer name is the cluster being curated, the names inside are subcluster IDs, and the values are the new labels. The special label `others` is used for any subcluster not listed explicitly.

Then `curate_clusters()` applies those manual labels and stores the result in `celltype_curated`. The final curated UMAP is saved as `results/05_curation/umap_curated.pdf`. This block is not executed automatically because the label choices depend on inspecting the figures.

R

```r
output_dir <- dir_05
Idents(ath_sc) <- "celltype"

# 1. Inspect a cluster (saves the combined subcluster + marker figure)
inspect_subcluster_markers(
  ath_sc, cluster_id = "Mesophyll",
  marker_table = marker_table, output_dir = output_dir
)

# 2. Map each cluster's subclusters to labels ("others" = any ID not listed),
#    then curate them in one call. Here Mesophyll's subclusters all map back
#    to "Mesophyll" (no relabeling), which is the pattern to use when the
#    subcluster inspection confirms the original label was already correct:
reassign <- list(
  "Mesophyll" = c("others" = "Mesophyll")
)
ath_sc <- curate_clusters(
  ath_sc, reassign,
  marker_table = marker_table, output_dir = output_dir
)

# 3. Save the curated annotation UMAP
save_pdf(
  DimPlot(ath_sc, group.by = "celltype_curated",
          label = TRUE, repel = TRUE, raster = FALSE),
  "umap_curated.pdf"
)

message("\nOK SUPPLEMENTARY S3 COMPLETE: curation complete")
```

Representative output from Supplementary Section S3 — subcluster inspection of the Mesophyll population (UMAP of its subclusters plus marker-gene panels used to decide whether to keep or split the label):

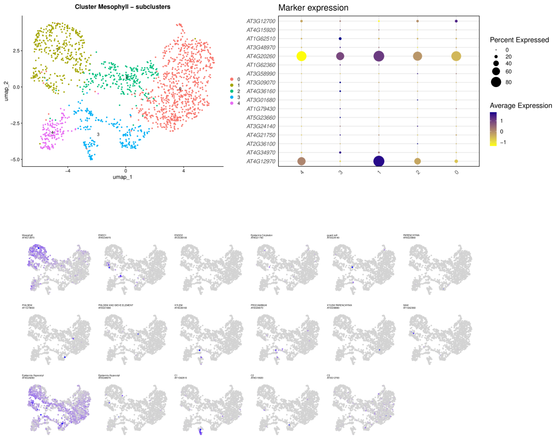


Figure 16: Mesophyll subcluster inspection: subcluster UMAP, marker dotplot, and per-gene marker panels.

In this example the Mesophyll subclusters are mapped back to `Mesophyll` *(no relabeling), which shows the inspect-then-curate mechanism without changing any label. The main workflow does not need this step: it continues with the labels from Section 7.*

Supplementary Table — Software dependencies and versions

Package versions are those bundled in the `psblab/scrnaseq:latest` Docker image (the public image used for all R and Python steps). R **base-priority** packages are omitted. Cell Ranger is 10x Genomics proprietary software run *outside* the container (Part 1); its version is fixed by the workflow.

Interpreters and command-line tools

Tool	Version
Cell Ranger	9.0.1
R	4.5.3
Python	3.12.3

Installing tmux or screen

OS	tmux	screen
Linux (Debian/Ubuntu)	`sudo apt install tmux`	`sudo apt install screen`
Linux (RHEL/CentOS/Fedora)	`sudo yum install tmux`	`sudo yum install screen`
macOS (Homebrew)	`brew install tmux`	ships with macOS by default
Windows	not natively available — see note below	not natively available — see note below

Windows users: this whole pipeline (Cell Ranger, Docker/Singularity, R, Python) is Linux-based. The practical options on Windows are WSL2 (https://learn.microsoft.com/windows/wsl/install) running Ubuntu locally, or an SSH client (Windows Terminal, PuTTY, or WSL2's own `ssh`) connecting out to a Linux server or HPC cluster where the rest of this guide applies directly. Once inside WSL2 or an SSH session on Linux, use the Linux commands above.

Installing a container runtime (Docker or Singularity/Apptainer)

Pick one; both run the same `psblab/scrnaseq:latest` image (see "Container startup" in Part 1).

Docker

OS	Install
Linux	https://docs.docker.com/engine/install/ (distro-specific packages)
macOS	Docker Desktop: https://docs.docker.com/desktop/install/mac-install/
Windows	Docker Desktop with the WSL2 backend: https://docs.docker.com/desktop/install/windows-install/

Singularity / Apptainer (no root required; common on HPC clusters where Docker is not permitted)

OS	Install
Linux	https://apptainer.org/docs/admin/main/installation.html
macOS / Windows	not natively supported — use a Linux VM, WSL2, or the remote server/cluster itself

On a shared HPC cluster, Singularity/Apptainer is usually already installed system-wide — check with `singularity --version` or `apptainer --version` before installing anything.

Troubleshooting Singularity/Apptainer

Symptom. `singularity build` finishes, but the startup command (`singularity exec ... scrnaseq.sif /bin/bash`) fails with `squashfuse not found` followed by `Operation not permitted` during extraction.

Cause. A `.sif` file is a compressed (squashfs) filesystem that must be mounted to run. Where Apptainer is installed with its setuid helper (common on shared clusters), this works with no further setup. In a rootless installation without that helper, Apptainer mounts the image through FUSE using `squashfuse`; if it is missing, Apptainer falls back to extracting the image, which fails when user namespaces are disabled or restricted, or when the temporary directory is unusable.

Prerequisite check. Run these three commands:

Bash

```bash
apptainer --version                        # must print a version number
which squashfuse fusermount                # must print a path for both programs
cat /proc/sys/user/max_user_namespaces     # must print a number greater than 0
```

The first command must always work. If Apptainer was installed with its setuid helper (common on shared clusters), nothing else is needed. Otherwise the second command must print a path for both programs, and the third must print a number greater than 0. If either is missing, the error above is expected.

Fixes, in order of effort.

1. Install `squashfuse` and `fuse` (on a cluster, ask the administrators).
2. Point Apptainer to a local disk with free space that is not mounted `noexec`, then retry:

Bash

```bash
export APPTAINER_TMPDIR=/path/to/local/disk/tmp
export APPTAINER_CACHEDIR=/path/to/local/disk/cache
mkdir -p "$APPTAINER_TMPDIR" "$APPTAINER_CACHEDIR"
```

3. Avoid mounting by building a directory instead of a `.sif`, and run that:

Bash

```bash
apptainer build --sandbox scrnaseq/ docker://psblab/scrnaseq:latest
apptainer exec --bind $(pwd):/workspace --pwd /workspace scrnaseq/ /bin/bash
```

4. If none of these work, contact your system administrators, or use Docker (Option B) on a machine where it is permitted.

On Ubuntu 24.04 and similar distributions, AppArmor can restrict unprivileged user namespaces by default; this requires an administrator to allow Apptainer.

R packages (421)

Package	Version	Package	Version
`abind`	1.4-8	`locfit`	1.5-9.12
`anndataR`	1.0.2	`lubridate`	1.9.5
`AnnotationDbi`	1.72.0	`magrittr`	2.0.5

Package	Version	Package	Version
`ape`	5.8-1	`maps`	3.4.3
`aplot`	0.2.9	`MASS`	7.3-65
`askpass`	1.2.1	`Matrix`	1.7-5
`assertthat`	0.2.1	`MatrixGenerics`	1.22.0
`assorthead`	1.4.0	`MatrixModels`	0.5-4
`backports`	1.5.1	`matrixStats`	1.5.0
`base64enc`	0.1-6	`memoise`	2.0.1
`basilisk`	1.22.0	`mgcv`	1.9-4
`batchelor`	1.26.0	`microbenchmark`	1.5.0
`beachmat`	2.26.0	`mime`	0.13
`beeswarm`	0.4.0	`miniUI`	0.1.2
`BH`	1.90.0-1	`minqa`	1.2.8
`biglm`	0.9-3	`modelr`	0.1.11
`Biobase`	2.70.0	`monocle3`	1.4.27
`BiocGenerics`	0.56.0	`nlme`	3.1-169
`BiocManager`	1.30.27	`nloptr`	2.2.1
`BiocNeighbors`	2.4.0	`nnet`	7.3-20
`BiocParallel`	1.44.0	`numDeriv`	2016.8-1.1
`BiocSingular`	1.26.1	`openssl`	2.4.1
`BiocVersion`	3.22.0	`org.At.tair.db`	3.22.0
`Biostrings`	2.78.0	`otel`	0.2.0
`bit`	4.6.0	`pak`	0.9.5
`bit64`	4.8.2	`parallelly`	1.47.0
`bitops`	1.0-9	`patchwork`	1.3.2
`blob`	1.3.0	`pbapply`	1.7-4
`boot`	1.3-32	`pbkrtest`	0.5.5
`BPCells`	0.3.1	`pbmcapply`	1.5.1
`brew`	1.0-10	`pheatmap`	1.0.13
`brio`	1.1.5	`pillar`	1.11.1
`broom`	1.0.13	`pkgbuild`	1.4.8
`bslib`	0.11.0	`pkgconfig`	2.0.3
`cachem`	1.1.0	`pkgdown`	2.2.0
`Cairo`	1.7-0	`pkgload`	1.5.2
`callr`	3.7.6	`plotly`	4.12.0
`car`	3.1-5	`plyr`	1.8.9
`carData`	3.0-6	`png`	0.1-9
`caTools`	1.18.3	`polyclip`	1.10-7
`cellranger`	1.1.0	`polynom`	1.4-1
`checkmate`	2.3.4	`praise`	1.0.0
`circlize`	0.4.18	`preprocessCore`	1.72.0
`class`	7.3-23	`prettyunits`	1.2.0
`classInt`	0.4-11	`processx`	3.9.0
`cli`	3.6.6	`profvis`	0.4.0
`clipr`	0.8.1	`progress`	1.2.3
`clue`	0.3-68	`progressr`	0.19.0
`cluster`	2.1.8.2	`promises`	1.5.0
`clusterProfiler`	4.18.4	`proxy`	0.4-29
`clustree`	0.5.1	`ps`	1.9.3
`codetools`	0.2-20	`pscl`	1.5.9
`colorspace`	2.1-2	`purrr`	1.2.2
`commonmark`	2.0.0	`quadprog`	1.5-8
`ComplexHeatmap`	2.26.1	`quantreg`	6.1
`conflicted`	1.2.0	`qvalue`	2.42.0
`corrplot`	0.95	`R.methodsS3`	1.8.2
`cowplot`	1.2.0	`R.oo`	1.27.1
`cpp11`	0.5.5	`R.utils`	2.13.0
`crayon`	1.5.3	`R6`	2.6.1
`credentials`	2.0.3	`ragg`	1.5.2
`crosstalk`	1.2.2	`RANN`	2.6.2
`curl`	7.1.0	`rappdirs`	0.3.4
`data.table`	1.18.4	`rbibutils`	2.4.1
`DBI`	1.3.0	`rcmdcheck`	1.4.0
`dbplyr`	2.5.2	`RColorBrewer`	1.1-3
`DelayedArray`	0.36.1	`Rcpp`	1.1.1-1.1
`DelayedMatrixStats`	1.32.0	`RcppAnnoy`	0.0.23
`deldir`	2.0-4	`RcppArmadillo`	15.2.7-1

Package	Version	Package	Version
`Deriv`	4.2.0	`RcppEigen`	0.3.4.0.2
`desc`	1.4.3	`RcppHNSW`	0.7.0
`DESeq2`	1.50.2	`RcppML`	0.3.7.1
`devtools`	2.5.2	`RcppProgress`	0.4.2
`diffobj`	0.3.6	`RcppRoll`	0.3.2
`digest`	0.6.39	`RcppTOML`	0.2.3
`dir.expiry`	1.18.0	`Rdpack`	2.6.6
`distributional`	0.7.0	`readr`	2.2.0
`doBy`	4.7.1	`readxl`	1.5.0
`docopt`	0.7.2	`reformulas`	0.4.4
`doParallel`	1.0.17	`rematch`	2.0.0
`DOSE`	4.4.0	`rematch2`	2.1.2
`dotCall64`	1.2	`remotes`	2.5.0
`DoubletFinder`	2.0.6	`reprex`	2.1.1
`downlit`	0.4.5	`reshape2`	1.4.5
`dplyr`	1.2.1	`ResidualMatrix`	1.20.0
`dqrng`	0.4.1	`reticulate`	1.46.0
`dtplyr`	1.3.3	`rhdf5`	2.54.1
`dynamicTreeCut`	1.63-1	`rhdf5filters`	1.22.0
`e1071`	1.7-17	`Rhdf5lib`	1.32.0
`ellipsis`	0.3.3	`RhpcBLASctl`	0.23-42
`enrichplot`	1.30.5	`Rhtslib`	3.6.0
`enrichR`	3.4	`rjson`	0.2.23
`eulerr`	8.0.0	`rlang`	1.2.0
`evaluate`	1.0.5	`rmarkdown`	2.31
`fansi`	1.0.7	`ROCR`	1.0-12
`farver`	2.1.2	`roxygen2`	8.0.0
`fastcluster`	1.3.0	`rpart`	4.1.24
`fastDummies`	1.7.6	`rprojroot`	2.1.1
`fastmap`	1.2.0	`rsample`	1.3.2
`fastmatch`	1.1-8	`Rsamtools`	2.26.0
`fgsea`	1.36.2	`RSpectra`	0.16-2
`fields`	17.3	`RSQLite`	3.53.1
`filelock`	1.0.3	`rstatix`	0.7.3
`fitdistrplus`	1.2-6	`rstudioapi`	0.18.0
`FNN`	1.1.4.1	`rsvd`	1.0.5
`fontawesome`	0.5.3	`Rtsne`	0.17
`fontBitstreamVera`	0.1.1	`rversions`	3.0.0
`fontLiberation`	0.1.0	`rvest`	1.0.5
`fontquiver`	0.2.1	`s2`	1.1.11
`forcats`	1.0.1	`S4Arrays`	1.10.1
`foreach`	1.5.2	`S4Vectors`	0.48.1
`forecast`	9.0.2	`S7`	0.2.2
`foreign`	0.8-91	`sass`	0.4.10
`formatR`	1.14	`ScaledMatrix`	1.18.0
`Formula`	1.2-5	`scales`	1.4.0
`fracdiff`	1.5-4	`scater`	1.38.1
`fs`	2.1.0	`scattermore`	1.2
`furrr`	0.4.0	`scatterpie`	0.2.6
`futile.logger`	1.4.9	`sctransform`	0.4.3
`futile.options`	1.0.1	`scuttle`	1.20.0
`future`	1.70.0	`selectr`	0.5-1
`future.apply`	1.20.2	`Seqinfo`	1.0.0
`gargle`	1.6.1	`sessioninfo`	1.2.3
`gdtools`	0.5.1	`Seurat`	5.5.0
`GeneOverlap`	1.46.0	`SeuratDisk`	0.0.0.9021
`generics`	0.1.4	`SeuratObject`	5.4.0
`GENIE3`	1.32.0	`SeuratWrappers`	0.4.0
`GenomeInfoDb`	1.46.2	`sf`	1.1-1
`GenomicRanges`	1.62.1	`shape`	1.4.6.1
`gert`	2.3.1	`shiny`	1.13.0
`GetoptLong`	1.1.1	`SingleCellExperiment`	1.32.0
`ggbeeswarm`	0.7.3	`sitmo`	2.0.2
`ggdist`	3.3.3	`slam`	0.1-55
`ggforce`	0.5.0	`slider`	0.3.3
`ggfun`	0.2.0	`snow`	0.4-4

Package	Version	Package	Version
ggiraph	0.9.6	sourcetools	0.1.7-2
ggnewscale	0.5.2	sp	2.2-1
ggplot2	4.0.3	spam	2.11-4
ggplotify	0.1.3	SparseArray	1.10.10
ggpubr	0.6.3	SparseM	1.84-2
ggraph	2.2.2	sparseMatrixStats	1.22.0
ggrastr	1.0.2	spatial	7.3-18
ggrepel	0.9.8	spatstat.data	3.1-9
ggridges	0.5.7	spatstat.explore	3.8-1
ggsci	5.0.0	spatstat.geom	3.8-1
ggsignif	0.6.4	spatstat.random	3.5-0
ggtangle	0.1.2	spatstat.sparse	3.2-0
ggtree	4.0.5	spatstat.univar	3.2-0
ggvenn	0.1.19	spatstat.utils	3.2-3
gh	1.6.0	spData	2.3.5
gitcreds	0.1.2	spdep	1.4-2
GlobalOptions	0.1.4	speedglm	0.3-4
globals	0.19.1	statmod	1.5.2
glue	1.8.1	stringi	1.8.7
GO.db	3.22.0	stringr	1.6.0
goftest	1.2-3	SummarizedExperiment	1.40.0
googledrive	2.1.2	survival	3.8-6
googlesheets4	1.1.2	svglite	2.2.2
GOSemSim	2.36.0	sys	3.4.3
gplots	3.3.0	systemfonts	1.3.2
graphlayouts	1.2.3	tensor	1.5.1
gridExtra	2.3	tester	0.3.0
gridGraphics	0.5-1	testthat	3.3.2
grr	0.9.5	textshaping	1.0.5
gson	0.1.0	tibble	3.3.1
gtable	0.3.6	tidydr	0.0.6
gtools	3.9.5	tidygraph	1.3.1
h5mread	1.2.1	tidyr	1.3.2
harmony	2.0.3	tidyselect	1.2.1
haven	2.5.5	tidytree	0.4.7
HDF5Array	1.38.0	tidyverse	2.0.0
hdf5r	1.3.12	timechange	0.4.0
hdWGCNA	0.4.11	timeDate	4052.112
here	1.0.2	tinytex	0.59
hexbin	1.28.5	treeio	1.34.0
highr	0.12	tweenr	2.0.3
Hmisc	5.2-5	tzdb	0.5.0
hms	1.1.4	UCell	2.14.0
htmlTable	2.5.0	UCSC.utils	1.6.1
htmltools	0.5.9	units	1.0-1
htmlwidgets	1.6.4	UpSetR	1.4.1
httpuv	1.6.17	urca	1.3-4
httr	1.4.8	urlchecker	1.0.1
httr2	1.2.2	usethis	3.2.1
ica	1.0-3	utf8	1.2.6
ids	1.0.1	uuid	1.2-2
igraph	2.3.2	uwot	0.2.4
impute	1.84.0	vctrs	0.7.3
ini	0.3.1	VennDiagram	1.8.2
IRanges	2.44.0	vipor	0.4.7
irlba	2.3.7	viridis	0.6.5
isoband	0.3.0	viridisLite	0.4.3
iterators	1.0.14	vroom	1.7.1
jquerylib	0.1.4	waldo	0.6.2
jsonlite	2.0.0	warp	0.2.3
kableExtra	1.4.0	WGCNA	1.74
KEGGREST	1.50.0	whisker	0.4.1
KernSmooth	2.23-26	withr	3.0.2
knitr	1.51	wk	0.9.5
labeling	0.4.3	WriteXLS	6.8.0
lambda.r	1.2.4	xfun	0.58

Package	Version	Package	Version
`later`	1.4.8	`xml2`	1.5.2
`lattice`	0.22-9	`xopen`	1.0.1
`lazyeval`	0.2.3	`xtable`	1.8-8
`leidenbase`	0.1.37	`XVector`	0.50.0
`lifecycle`	1.0.5	`yaml`	2.3.12
`limma`	3.66.0	`yulab.utils`	0.2.4
`listenv`	0.10.1	`zellkonverter`	1.20.1
`littler`	0.3.23	`zip`	2.3.3
`lme4`	2.0-1	`zoo`	1.8-15
`lmtest`	0.9-40		

Python packages (207)

Package	Version	Package	Version
`adjustText`	1.3.0	`notebook`	6.5.7
`anndata`	0.12.16	`notebook_shim`	0.2.4
`annotated-types`	0.7.0	`numba`	0.65.1
`anyio`	4.13.0	`numcodecs`	0.16.5
`argon2-cffi`	25.1.0	`numexpr`	2.14.1
`argon2-cffi-bindings`	25.1.0	`numpy`	2.4.6
`array-api-compat`	1.14.0	`numpy-groupies`	0.11.3
`arrow`	1.4.0	`nvidia-cublas`	13.1.1.3
`asttokens`	3.0.1	`nvidia-cuda-cupti`	13.0.85
`async-lru`	2.3.0	`nvidia-cuda-nvrtc`	13.0.88
`attrs`	26.1.0	`nvidia-cuda-runtime`	13.0.96
`babel`	2.18.0	`nvidia-cudnn-cu13`	9.20.0.48
`beautifulsoup4`	4.14.3	`nvidia-cufft`	12.0.0.61
`bleach`	6.3.0	`nvidia-cufile`	1.15.1.6
`blosc2`	4.4.1	`nvidia-curand`	10.4.0.35
`cellbender`	0.3.2	`nvidia-cusolver`	12.0.4.66
`certifi`	2026.5.20	`nvidia-cusparse`	12.6.3.3
`cffi`	2.0.0	`nvidia-cusparselt-cu13`	0.8.1
`charset-normalizer`	3.4.7	`nvidia-nccl-cu13`	2.29.7
`click`	8.4.1	`nvidia-nvjitlink`	13.0.88
`comm`	0.2.3	`nvidia-nvshmem-cu13`	3.4.5
`contourpy`	1.3.3	`nvidia-nvtx`	13.0.85
`cuda-bindings`	13.3.1	`opt_einsum`	3.4.0
`cuda-pathfinder`	1.5.5	`packaging`	26.2
`cuda-toolkit`	13.0.2	`palantir`	1.4.4
`cycler`	0.12.1	`pandas`	2.3.3
`debugpy`	1.8.21	`pandocfilters`	1.5.1
`decorator`	5.3.1	`parso`	0.8.7
`defusedxml`	0.7.1	`patsy`	1.0.2
`donfig`	0.8.1.post1	`pexpect`	4.9.0
`elpigraph-python`	0.3.4	`pillow`	12.2.0
`entrypoints`	0.4	`pip`	26.1.2
`executing`	2.2.1	`platformdirs`	4.10.0
`fa2_modified`	0.4	`plotly`	6.7.0
`fast-array-utils`	1.4.1	`prometheus_client`	0.25.0
`fastjsonschema`	2.21.2	`prompt_toolkit`	3.0.52
`filelock`	3.29.0	`psutil`	7.2.2
`fonttools`	4.63.0	`ptyprocess`	0.7.0
`fqdn`	1.5.1	`pure_eval`	0.2.3
`fsspec`	2026.4.0	`py-cpuinfo`	9.0.0
`google-crc32c`	1.8.0	`pycparser`	3.0
`h11`	0.16.0	`pydantic`	2.13.4
`h5py`	3.16.0	`pydantic_core`	2.46.4
`httpcore`	1.0.9	`Pygments`	2.20.0
`httpx`	0.28.1	`pynndescent`	0.6.0
`idna`	3.18	`pyparsing`	3.3.2
`igraph`	1.0.0	`pyro-api`	0.1.2
`ipykernel`	6.29.5	`pyro-ppl`	1.9.1
`ipython`	9.14.0	`python-dateutil`	2.9.0.post0

Package	Version	Package	Version
`ipython-genutils`	0.2.0	`python-igraph`	1.0.0
`ipython_pygments_lexers`	1.1.1	`python-json-logger`	4.1.0
`ipywidgets`	8.1.8	`pytz`	2026.2
`isoduration`	20.11.0	`PyYAML`	6.0.3
`jax`	0.10.1	`pyzmq`	27.1.0
`jaxlib`	0.10.1	`referencing`	0.37.0
`jaxopt`	0.8.5	`requests`	2.34.2
`jedi`	0.20.0	`rfc3339-validator`	0.1.4
`Jinja2`	3.1.6	`rfc3986-validator`	0.1.1
`joblib`	1.5.3	`rfc3987-syntax`	1.1.0
`json5`	0.14.0	`rpds-py`	2026.5.1
`jsonpointer`	3.1.1	`rpy2`	3.6.7
`jsonschema`	4.26.0	`rpy2-rinterface`	3.6.6
`jsonschema-specifications`	2025.9.1	`rpy2-robjects`	3.6.5
`jupyter`	1.1.1	`scanpy`	1.12.1
`jupyter-console`	6.6.3	`scFates`	1.2.4
`jupyter-events`	0.12.1	`scikit-learn`	1.9.0
`jupyter-highlight-selected-word`	0.2.0	`scikit-misc`	0.5.2
`jupyter-lsp`	2.3.1	`scipy`	1.17.1
`jupyter_client`	7.4.9	`scverse-misc`	0.0.7
`jupyter_contrib_core`	0.4.2	`seaborn`	0.13.2
`jupyter_contrib_nbextensions`	0.7.0	`Send2Trash`	2.1.0
`jupyter_core`	5.9.1	`session-info2`	0.4.1
`jupyter_nbextensions_configurator`	0.6.4	`setuptools`	81.0.0
`jupyter_server`	2.19.0	`shapely`	2.1.2
`jupyter_server_terminals`	0.5.4	`simpleppt`	1.1.7
`jupyterlab`	4.5.7	`six`	1.17.0
`jupyterlab_pygments`	0.3.0	`soupsieve`	2.8.4
`jupyterlab_server`	2.28.0	`stack-data`	0.6.3
`jupyterlab_widgets`	3.0.16	`statsmodels`	0.14.6
`kiwisolver`	1.5.0	`sympy`	1.14.0
`lark`	1.3.1	`tables`	3.11.1
`legacy-api-wrap`	1.5	`terminado`	0.18.1
`leidenalg`	0.12.0	`texttable`	1.7.0
`llvmlite`	0.47.0	`threadpoolctl`	3.6.0
`loompy`	3.0.8	`tinycss2`	1.5.1
`lxml`	6.1.1	`torch`	2.12.0
`lxml_html_clean`	0.4.5	`tornado`	6.5.6
`MarkupSafe`	3.0.3	`tqdm`	4.67.3
`matplotlib`	3.10.9	`traitlets`	5.15.0
`matplotlib-inline`	0.2.2	`triton`	3.7.0
`mellon`	1.7.1	`typing-inspection`	0.4.2
`mistune`	0.8.4	`typing_extensions`	4.15.0
`ml_dtypes`	0.5.4	`tzdata`	2026.2
`mpmath`	1.3.0	`tzlocal`	5.4.4
`msgpack`	1.1.2	`umap-learn`	0.5.12
`narwhals`	2.22.0	`uri-template`	1.3.0
`natsort`	8.4.0	`urllib3`	2.7.0
`nbclassic`	1.3.3	`wcwidth`	0.7.0
`nbclient`	0.10.4	`webcolors`	25.10.0
`nbconvert`	6.5.4	`webencodings`	0.5.1
`nbformat`	5.10.4	`websocket-client`	1.9.0
`ndindex`	1.10.1	`widgetsnbextension`	4.0.15
`nest-asyncio`	1.6.0	`zarr`	3.2.1
`networkx`	3.6.1		